Water Resources Research

RESEARCH ARTICLE

Hidden support share as a conditional risk screen for annual thermal extremes in incomplete river-temperature records

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Key Points:

- Under a 5% false-admit cap set on calibration data only, risk ranking keeps 27 points more records than a gap-length rule
- Engineered peak-overlap adds nothing measurable to a same-capacity learner, so hidden support is conditional, not necessary
- A sealed 49-network panel cuts designed-outage errors by 5.9 points; on the gaps networks actually keep the gain is smaller

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Abstract Thermal archives are screened by completeness thresholds, assuming that how much data is missing decides whether an annual statistic can be trusted. That assumption has not been tested against the statistic's own support, because a gap's length and position cannot be varied independently in a record that failed on its own. Here we combine 5114 complete river station-years from 146 networks with 51524 designed outages, vary a gap's length and position separately, and score admission rules on a retrospective holdout and on an external panel of 49 networks. An annual hottest 30-day mean should be admitted on indicator-conditioned risk rather than uniform completeness: a fitted risk score reading only decision-time information beats the gap-length rule by 6.2 points at 70% retention ([−7.2, −5.4]), and under a five-per-cent false-admit cap whose thresholds are chosen on calibration networks alone it admits 27 points more of the candidate pool than the gap-length rule ([23.9, 30.3]). Engineered peak-overlap features add nothing measurable to a same-capacity learner (+0.05 points, [−0.17, +0.23]), so hidden support is a useful conditional predictor but not a necessary mechanism. Crossed controls show that hiding the peak window and lengthening the gap both raise the error, by amounts the reconstructor sets. On the external panel the frozen certificate screen cuts designed-outage errors by 5.9 points ([−6.9, −4.9]); on the gaps these networks actually keep, the gain is about 0.8 points and must not be pooled into an operational guarantee. Completeness thresholds remain the right axis for annual means and the wrong default for extremes.

Plain Language Summary Rivers are getting warmer, and managers ask how hot a river got last summer. The answer needs a year of daily temperatures, but records have holes, and the usual screen counts how many days are missing. We show that this is the wrong question for a summer peak: what matters is whether the missing days include the few weeks the peak depends on. Using complete years from 146 river networks, we place gaps of identical length in different positions and compare what two statistics lose. A gap over the peak window destroys the hottest 30-day mean while the same gap in autumn barely moves it; a whole-year average behaves in the opposite way. A rule that screens on this "hidden support" instead of on gap length removes most large errors on a retrospective holdout, but when we look at the gaps these networks really experience the gain shrinks to a fraction. An independent multi-provider confirmation under repaired data QA recovers a −5.9-point screening gain. Results about exact crossover days, autumn maintenance benefits and "peak never observed" counts are exploratory or use proxies and should not yet guide operations.

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1. Introduction

Cold-water habitat assessment, thermal-regime classification and criterion-exceedance accounting all reduce a year of daily water temperature to one number — a hottest 30-day mean, an annual mean, a seasonal amplitude (Caissie, 2006; Isaak et al., 2017) — and it is that number, not the daily series, that enters a management judgement. The error that matters is therefore the error of the derived statistic. On the panel assembled here, computing the summer peak from recorded days alone, which is the default once a station-year passes a completeness screen, carries a mean absolute error of **1.305 °C**, or 1.23 year-to-year standard deviations of the quantity itself, and 24.9% of station-years are wrong by more than 1.0 °C. An error of that size turns a thermally stressful year into an unremarkable one.

Current practice assumes that the amount of missing data decides whether the derived number may be used, and two families of rule encode that assumption (Table 1). Collection standards specify what must be present: Mauger, Shaftel, Trammell, Geist, and Bogan (2015), writing minimum standards for a state whose records feed regional analyses, require a minimum sampling duration of one calendar month and recommend a deployment period of 1 June to 31 August "to capture the thermal maxima"; the same standards govern

which of those records enter the regional assessment (Mauger et al., 2017). Quantitative guidance constrains what may be absent: Johnson et al. (2021) report that "datasets with less than seven weeks of consecutively missing data enabled estimation of all annual signal parameters with reasonable accuracy", expanding to "approximately 20 weeks" when missing data are imputed. The gap-filling literature optimises the accuracy of the filled series rather than the admissibility of the completed year (Mohseni et al., 1998; Li et al., 2017), and the closest prior work fills incomplete summer records so that a thermal peak can be computed, applies that filling uniformly, declines no station-year, and admits July–August-only records on the climatological position of the hottest window (Beaufort et al., 2022). Large-sample thermal models optimise daily or reach-scale prediction skill (Isaak et al., 2017; Corona & Hogue, 2025; Kratzert et al., 2023), and selective-prediction methods control error at a fixed rejection rate (Geifman & El-Yaniv, 2017; Zaoui et al., 2020). None of these rules measures how much of the target statistic's own support a record contains.

The assumption is plausible for some statistics and wrong for others. An annual mean is a sum over days, so removing k days removes k terms and it matters little which. A hottest 30-day mean is an extreme supported on a short window: if the outage misses that window the statistic is recoverable almost exactly however long the outage is, and if the outage covers it the statistic must be inferred however short the outage is. The direction of a completeness rule's error is therefore opposite in the two cases, and an average over a mixed archive conceals the reversal. The literature is consistent with that reversal without having isolated it: reported timing effects on annual thermal parameters are inconsistent across regions, years and parameters (Johnson et al., 2021); reconstruction bias depends on where the target window falls (Beaufort et al., 2022); systematic, time-varying instrument bias rather than random noise dominates long-term monitoring error (Jackson et al., 2025); and statistic-aware completeness conventions do exist in climate monitoring, but they are defined by counts or seasons rather than by the statistic's support (Zhang et al., 2011). Irregular sampling and sampling artifacts likewise bias means and trends and decide which model structures remain trustworthy (Grey et al., 2024), and reconstruction frameworks have been built for regulated and data-sparse basins, including Chinese catchments and the Yangtze mainstem (Huang et al., 2023; Wang, 2025); those studies answer estimation and reconstruction questions, not the archive-admission question for a named annual extreme.

Testing the assumption requires separating two properties of a gap that real records entangle. Gap length and the share of the target window that is hidden correlate at −0.86 in our own panel, so a comparison across real gaps cannot say whether a rule works because it avoids long gaps, because it avoids badly placed ones, or because both simply track how much record the station has. A second obstacle is truth: in exactly the years where the admission decision matters, the target statistic is unknown by construction, so the benefit of a rule cannot be measured where it would be applied. A third is that evaluations report a rule's discrimination rather than the composition of the archive it leaves behind, and composition is what decides which catchments remain available for analysis.

To address these limitations we assemble 5114 complete river station-years from 146 networks and nine national agencies, place 51524 outages on complete years so that the two properties of a gap can be varied independently, and score admission rules on held-out river networks, on sealed panels that took no part in development, and — without any truth — on the 2422 records in this same archive that really do have gaps. Three questions organise the paper, and the last matters most, because a screening rule is judged by the archive it leaves behind. First, which property of a gap predicts the error of a derived thermal statistic, and does the answer depend on which statistic is being computed? Second, is that dependence a dichotomy between two regimes or a continuous transition, where does it turn over, and how much of the gain survives the gap calendar these networks actually keep? Third, what does conditioning on it change for the error among retained station-years, for which stations survive in the archive, and for the undecided records?

2. Study system, data and identification design

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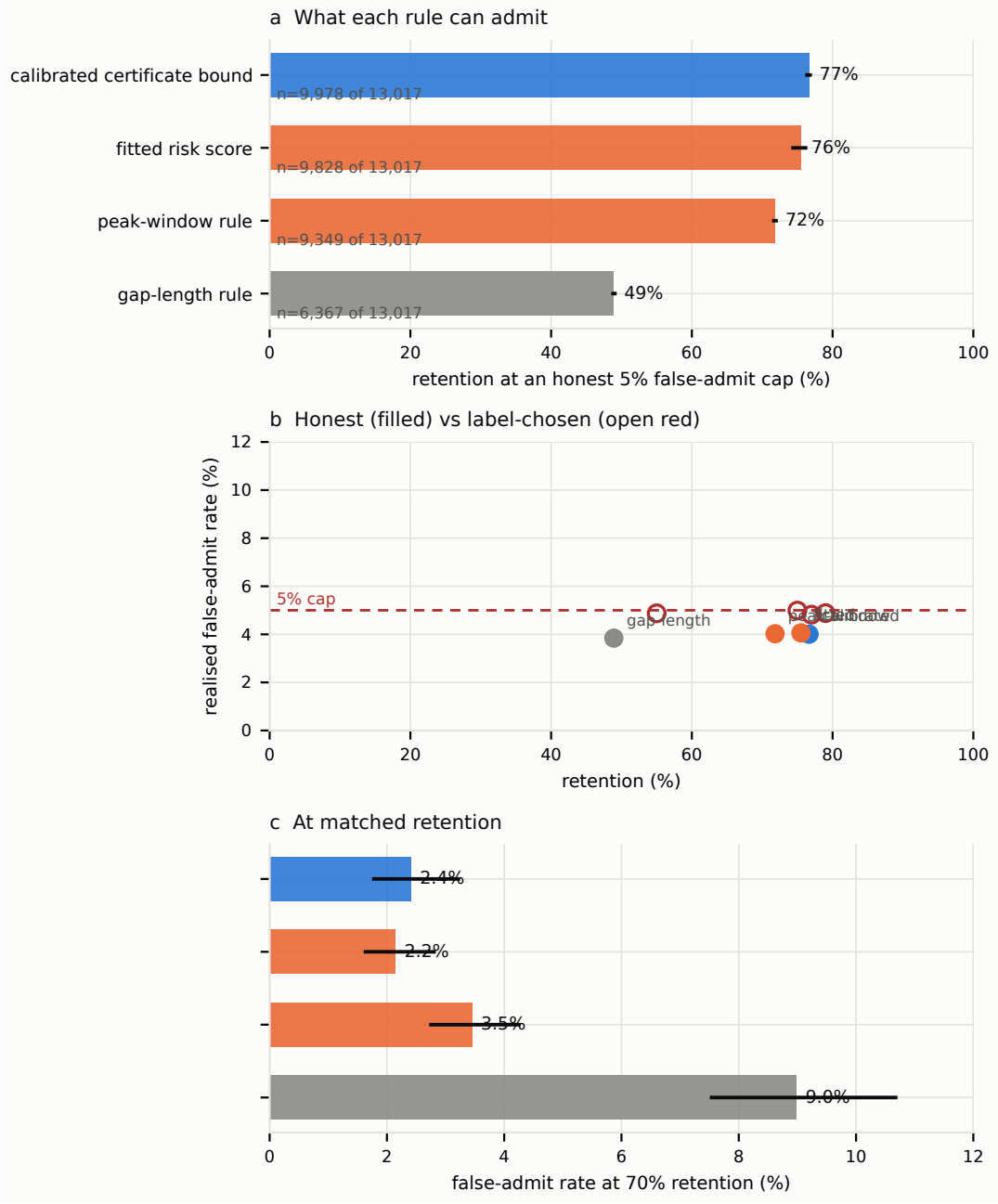


Figure 1: What the admission rule changes, on the 49-network external panel (13 017 designed-outage cases; every threshold chosen on calibration networks only, so no panel label is read). (a) Retention at an honest 5% false-admit cap: the calibrated certificate bound admits 77% of candidates, the fitted risk score 76%, the transparent peak-window rule 72% and the gap-length rule 49%; bars are the mean over three network-fold seeds and the black line spans the seeds' range. (b) The same rules on the risk–retention plane, filled markers for the honest operating points and open red markers for the point the earlier draft quoted, which was chosen with the evaluation labels and is not deployable; the dashed line is the 5% cap. (c) The false-admit rate each rule leaves at a matched 70% retention, with network-clustered 95% intervals: 2.4% for the calibrated bound against 9.0% for the gap-length rule. Source data: `figures/wrr/f1_decision_data.json`.

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Table 1: What each body of work optimises, decides on, and conditions on. The rows are ordered by what they assume about the conditioning variable, not by date.

body of work	optimises	decision unit	conditioning axis
completeness guidance (Johnson et al., 2021)	tolerable amount of missing data	station-year	outage size (timing tested; effects reported inconsistent)
collection standards (Mauger et al., 2015)	comparability across a programme	deployment	season and duration
gap filling (Mohseni et al., 1998; Li et al., 2017)	accuracy of the filled series	day	air temperature, donors
peak reconstruction (Beaufort et al., 2022)	spatial coverage of the thermal peak	reach	air regression applied uniformly; climatological summer window used to admit partial-season records
large-sample models (Isaak et al., 2017; Corona & Hogue, 2025; Kratzert et al., 2023)	prediction skill	day or reach	meteorology, catchment attributes
selective prediction (Geifman & El-Yaniv, 2017; Zaoui et al., 2020)	error at a fixed rejection rate	case	model confidence
this work	**error of the derived statistic among retained years**	**station-year**	**share of the statistic's support hidden**

2.1. The statistic, its support, and the share a gap hides

Let a station-year be a sequence $y = (y_1, \dots, y_T)$ of daily mean water temperatures, $T \in \{365, 366\}$, and let $q(\cdot)$ be the derived statistic of interest. The family we sweep is the hottest consecutive k-day mean

$$q_k(y) = \max_t \frac{1}{k} \sum_{i=t}^{t+k-1} y_i, \tag{1}$$

which is the annual maximum at $k = 1$ and the annual mean at $k = T$; the coldest 30-day mean, the annual amplitude, and the sine-fit mean, phase and amplitude of Johnson et al. (2021) are scored alongside.

A gap is a binary mask $m \in \{0, 1\}^T$; the analyst observes the days where $m_t = 0$. A reconstruction $\hat{y} = R(y_{\bar{m}}, Z)$ completes the year using side information Z — synchronous donor stations, air temperature, the station's own history — and reports $q(\hat{y})$. The quantity that matters is the error of the reported statistic,

$$L = |q(\hat{y}) - q(y)|, \tag{2}$$

and a station-year is called *consequential* when $L > \tau$ for a tolerance τ in the units of q.

The property of the gap we propose as the governing one follows from how much each day moves the statistic. Writing $w_t^q = |\partial q(y)/\partial y_t|$ for the discrete influence function of q at y (Hampel, 1974), the *hidden mass* of a gap is the share of that influence it removes,

$$h_q(m) = \frac{\sum_{t:\, m_t = 1} w_t^q}{\sum_t w_t^q}. \tag{3}$$

For the annual mean every w_t is equal and h_q is the missing fraction; for the hottest k-day mean the influence is concentrated on the k days achieving the maximum and h_q is the share of that window the gap hides; for the sine-fit parameters the weights are smooth functions of day of year — a constant for the fitted mean, $|\sin(\theta_t - \phi)|$ for the amplitude, which is largest at both extremes, and $|\cos(\theta_t - \phi)|$ for the phase, which is largest a quarter-year from the peak. Equation (3) therefore contains the incumbent axis as a special case rather than opposing it, which is why the two regimes reported below are ends of one scale.

Equation (3) cannot be evaluated at decision time, because the weights depend on the days the gap hides. What an analyst has is the *climatological* support: the window the station's other complete years say the maximum should fall in. That estimate has an error, and Section 4.3.1 shows the error is large enough to change what a screening rule should condition on.

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2.2. Records, and what each dataset is for

Every result below is therefore reported with the measurement contract of the cases it rests on (`measurement_contract_table.csv`, `final_coverage_table.csv`). Of the 40 704 development cases, 74.4% come from providers that document a daily mean (84 networks), 15.5% are project aggregates of sub-daily or instantaneous values (14 networks, chiefly CHMI 10-minute and Rijkswaterstaat raw readings), 8.3% come from ten networks whose provider never states whether the published daily value is a mean (all ARSO), and 1.7% are the three Finnish morning reference readings. The reserved panel is similar; the external wave-2 panel is 94.4% documented daily means. Where a headline number is quoted for the pooled panel, the stratum table above gives the same contrast inside the documented-daily-mean subset, so the reader can see how much of it rests on records whose statistic is stated.

Daily water-temperature series were assembled from nine national providers: ARSO (Slovenia), CHMI (Czechia), FOEN (Switzerland), GKD Bayern and LUBW (Germany), NVE (Norway), Rijkswaterstaat (Netherlands), SYKE (Finland) and the USGS (United States). Acquisition contracts differ, and we audited each one against the provider's own documentation (`measurement_provenance_dictionary.csv`). Five providers document a daily mean and the parser reads it: USGS (`parameterCd=00010`, `statCd=00003`), FOEN (`data_1day_mean`), GKD Bayern (`Tageswerte`, column `Mittelwert`), NVE HydAPI (resolution 1440, documented as mean) and, for the transport panel below, the UK Environment Agency (measure period 86400, "daily mean Temperature"), with LUBW's `Tagesmittelwert` documented but its opaque workbook column only partially re-verified. Four providers do not meet that contract as read: CHMI's parser averages 10-minute *instantaneous* values (T_O) even though the provider also publishes a daily-mean product (T_D); Rijkswaterstaat's parser averages raw `NVT` readings even though a live 24 h mean parameter (`GEM24H`) exists; SYKE's public day values are morning nearshore reference readings (08:00, 20 cm below the surface) rather than 24 h means; and ARSO's daily value has no published mean-versus-hour statement. We therefore treat "daily mean" as a contract that holds for part of the panel and report the consequence: restricted to the five providers whose daily value is a documented provider daily mean, the development contrast is −5.7 points ([−6.5, −4.9], 84 networks) and the reserved contrast −6.6 ([−7.9, −5.3], 29 networks), against −6.4 and −7.1 for the full panel (`amendments/provider_contract_sensitivity/`). The sealed confirmation is USGS-only and so is entirely inside the documented stratum. The three SYKE stations remain flagged in `results/wrr_submission_ready/measurement_provenance_stations.csv` and are read as a sensitivity stratum, not as proven daily means. A *network* is one provider's group of stations on a common river system. It is the unit at which donor stations are available to one another, and it is the clustering unit for every interval reported here, because station-years within a network share weather, regulation and instrumentation.

Waterbody class was assigned per network and non-river objects — seas, lakes, canals, estuaries, deltas — are excluded rather than averaged into a river result. Reference series are never imputed: a calendar year with any missing day supplies no truth. Stations with fewer than four fully observed calendar years are excluded because the estimator protocol requires them. This leaves **146 networks, 457 stations and 5114 fully observed station-years**.

The same 457 stations contribute **2422 further station-years that are genuinely incomplete**. These carry no truth and are therefore not part of the scored population; they are the population a screening rule exists for. Section 4.4.4 reports overlap-rule decisions on that panel; the frozen certificate can also be applied through a truth-free `predict` interface (repository amendment `results/wrr_submission_ready/e2_certificate_predict/`).

Station coordinates were assembled for all 457 stations, from the providers' own catalogues, from the USGS NWIS site service for 195 United States gauges, and from ARSO's public station feed, and daily 2 m air temperature was joined from NASA POWER for the whole panel, 1981 to 2026 (NASA Langley Research Center, 2026). For the United States subset, basin characteristics were joined from GAGES-II (Falcone, 2011) on the eight-digit gauge identifier, matching **140 of 204** eligible stations across 57 networks. Basin attributes are used to ask which rivers are hard (Section 4.4.8) and which rivers a screen keeps (Section 4.4.7); they are deliberately not used inside the screening rule, because they are unavailable for the other eight agencies.

Table 2 lists every quantity the analysis uses, what it means physically, where its value comes from, and

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Table 2: Quantities used in the analysis.

quantity	physical meaning	source	calibrated
h_q, hidden mass	share of the statistic's influence removed by the gap	mask and the station's own complete years	no
peak-window overlap	h_q for a 30-day extreme, from the climatological window	mask and training years	no
warm-season coverage	share of May–September days observed	mask	no
longest run, missing fraction	size of the gap	mask	no
observed-peak margin	how far the observed days already constrain the annual maximum	observed days and training climatology	no
donor count, coverage, correlation	how much synchronous anomaly information neighbours carry	training-period deseasonalised series	no
K, ridge penalty of the donor and air estimators	strength of shrinkage toward the seasonal mean	leave-one-year-out within station	yes, within station
conformal scale	multiplicative factor turning a predicted bound into a valid one	calibration networks, disjoint from fitting and test	yes, on networks
τ, tolerance	consequential error in the units of the statistic	0.47 of the endpoint's own interannual SD; swept 0.25–1.0	no, declared
basin attributes	elevation, slope, baseflow index, drainage area	GAGES-II, United States subset	no, not an input

whether it was calibrated. Networks — never station-years — were split into development and reserved, stratified on provider and on donor regime, with the seed recorded before any scoring. This gives **110 development and 36 reserved networks**. That reserved panel is a retrospective network holdout for the frozen v17 screen: 28 of 36 networks have prior development, descriptive or scoring use in earlier programme stages (`results/wrr_submission_ready/prior_access_ledger_reserved_networks.csv`). It is not an unused confirmation set for analyses developed after opening. A sealed confirmation was therefore recruited externally: 39 USGS HUC4 networks (91 stations; reported HUC2 contexts need recount under string-safe HUC identifiers; 8973 designed outage cases) that lie outside the v17/v18 development panel were ingested, hashed, and scored once with the frozen certificate (`p0_07_sealed_confirmation/verdict.json`). That score is retained as an executed artifact, but the 2026-10-13 audit finds it cannot yet support a confirmation claim: USGS no-data sentinels (−999999) entered complete-year truth chains, and cases whose selected reconstructor returned no annual metric were dropped before the 70% retention cut, so the published operating point is not 70% of the candidate population (`AMENDMENT_claim_status_freeze_20261013.md`; amendments under `p0_02_qc_ledger/` and `p0_03_retention_denominator/`). Formal effect of that first sealed score: withdrawn after audit. A repo-wide prior-access diagnostic still applies (24 of 39 networks appear in earlier iterations). A pre-registered multi-agency transport test (SMHI, UK Environment Agency, Hub'Eau; 32 primary networks / 152 stations / 17 826 cases, with NVE only as a secondary stratum; manuscript text that mixed 245 extended-panel stations into the primary counts is corrected here) was likewise scored once, but its quality-checked gate incorrectly mixed the NVE secondary panel (≈ 97% of historical quality-checked cases) and fails when recomputed on the primary quality-checked subset alone (2 networks, 307 cases; clustered interval includes zero; `amendments/f03_primary_qc_diagnostic/`). That transport seal remains withdrawn. Independently, a wave-2 confirmation panel was recruited under a pre-sealed protocol (`P0_07_NEW_CONFIRMATION_SEAL.json`), excluding programme-used stations and leftover opened-wave ingest: 49 networks / 143 stations / 13 017 cases across USGS, Hub'Eau and UK EA. At 70% retention and 0.5 °C tolerance it yields $\Delta = -5.89$ points ($[-6.92, -4.91]$), zero reconstruction failures, USGS and Hub'Eau both negative, and the USGS quality-checked subset agreeing (`p0_07_wave2_ingest/p0_07_wave2_confirmation/verdict.json`). Section 2.2 reports the retrospective holdout that remains valid and labels the withdrawn first seals accordingly.

2.3. Gap design: two arms, and why both are needed

A case is one station-year under one gap realization. Two arms are used and reported separately, because they answer different questions and are not one population. The *empirical* arm samples real gap runs — length and start month — from the same network's own records and transplants them onto fully observed

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years, so that operational numbers reflect the geometry agencies experience. The *stratified* arm crosses gap length {14, 30, 60, 90, 120, 180, 270} days with the overlap between the gap and the station's climatological peak window, so that the response surface is estimable rather than accidental. The design yields **51524 cases**, 40704 development and 10820 reserved.

Both arms transplant a mask onto a year whose values were observed, which assumes the mask is uninformative about the values it hides. Section 4.4.9 tests that assumption against the archive rather than asserting it. Gaps are placed inside one calendar year; failures that straddle New Year damage two annual statistics while hiding neither warm season, and the design cannot represent that case.

Each endpoint is scored against a tolerance equal to the same fraction, 0.47, of *its own* interannual standard deviation at the same stations. For the annual peak that scale is 1.06 °C, giving a primary tolerance of 0.50 °C; the scale falls monotonically with support width, from 1.18 °C at $k = 1$ to 0.55 °C at $k = 365$. The fraction is swept at 0.25, 0.47 and 1.0, and no conclusion about which axis wins is claimed unless it is invariant across the sweep. The reference series carries instrument error of roughly 0.1 to 0.2 °C, so the primary tolerance is about three times sensor accuracy; Section 4.4.9 perturbs the reference by 0.15 °C and repeats the analysis.

3. Methods

Each choice below answers one of the obstacles in the introduction, and the order of the subsections follows that logic rather than the order in which the code runs: the reconstruction defines what a deployed rule may read; the rule defines the two candidate conditioning axes and the information barrier between them and the truth; the test protocol fixes the operating point, the clustered inference unit and the falsification tests declared before scoring.

3.1. Reconstruction, and what each estimator is allowed to see

Every case is completed by each supported estimator: the statistic computed from recorded days alone (current practice), a day-of-year median climatology, a smoothed climatology, linear interpolation where both flanks exist, ridge regression on the visible subset of synchronous donors, and an air-driven ridge with lagged air temperature and harmonics. Estimators are fit leave-one-year-out within the station, which is what offline archive repair permits; Section 4.4.9 repeats the study under the stricter protocol in which only prior years are available. Support is declared per method: donor regression does not read a flanking observation, so requiring one would discard usable cases, and absent donor columns are never imputed but the model is refit on the visible subset.

One weakness in that library was worth removing before the comparison was run. Air temperature was available for 2.3% of development cases in an earlier version of this analysis, so the reconstruction this field actually uses was effectively absent, and 36 of the 110 development networks have no synchronous donor at all. With coordinates joined for all 457 stations and NASA POWER pulled for the whole panel, air forcing now covers 98.6% of development cases and 100% of reserved cases. The retrospective reserved holdout is not re-scored against the new forcing; Section 4.4.3 reports it as a development-panel sensitivity analysis.

The reconstruction layer is not the contribution of this paper and is kept deliberately simple. Its role is to make the comparison between conditioning variables fair, which requires only that the estimator paired with the incumbent rule is the strongest deployable one; Section 4.4.3 tests that requirement directly rather than assuming it.

3.2. The screening rule, and the information it is forbidden to read

For each case the screen predicts an upper bound on the absolute error each supported estimator would produce, selects the estimator with the smallest bound, and emits a conformal interval on the statistic together with an admission decision at an analyst-chosen tolerance. Its inputs are the first six rows of Table 2. No

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input may read a day the gap hides, and that restriction is enforced in code rather than by convention.

Calibration is split conformal (Lei et al., 2018), and the risk-control framing follows the distribution-free line of Angelopoulos, Bates, Fisch, Lei, and Schuster (2022) with the caveat that selective observation is itself an existing problem statement (Bai et al., 2026); what is specific here is that selection happens among reconstructions chosen per station-year. Two properties matter. It is *selection-conditional*, because taking the smallest of several bounds is a winner's curse that per-estimator calibration cannot see, and *multiplicative*, because the error scale spans an order of magnitude between a gap that misses the peak window and one that covers it. Fitting, calibration and test networks are always disjoint.

3.3. Test protocol: fixed operating point, clustered unit, declared falsifications

Before any rule was fit we measured, on development cases, how much a rule with perfect foresight could win inside each candidate action space, and compared it with the margin the reserved panel can resolve. Two natural framings — adopt or reject a fixed reconstruction, and choose among reconstruction methods — have ceilings of +0.038 and +0.080 °C against a threshold of 0.136 °C, and were not pursued. Retain-or-exclude clears its gate by a factor of five and is the action space studied here. A decision experiment whose ceiling is below its own resolvable margin cannot produce evidence in either direction, and running one consumes a reserved panel for nothing.

Every interval resamples whole networks, because station-years within a network are not independent and reusing a station-year under several gap realizations does not create independent observations. The pre-registered primary criterion, frozen before the reserved panel was opened, is that at the retention of the matched length rule the screen must reduce the consequential-error rate by at least five percentage points with a network-clustered interval excluding zero. Precision was computed in advance: the reserved panel resolves 1.44 points, so the declared effect is 3.5 times the resolution.

Where several endpoints are compared, the direction of each contrast was predicted in advance from the form of the statistic, which is the defence against multiplicity. Three of those predictions were not met, and Sections 4.3.1 and 4.2.2 report them as failures rather than reshaping them into successes.

4. Results

4.1. The phenomenon: the same gap damages different annual statistics differently

Whether an incomplete year can carry a summer extreme depends strongly on where its gap falls. One real record shows the two ends of the range: a 30-day gap over the peak window costs 3.99 °C in the hottest 30-day mean, while the same 30 days in late autumn cost nothing; the annual mean on this example moves by −1.03 and +0.24 °C and therefore does *not* reverse the order (Figure 2). Across the development panel the relative importance of timing versus length still differs by endpoint, which is enough to change which station-years an archive should keep, without requiring a universal "position dominates length" law.

4.1.1. Computing the peak from recorded days alone costs more than a year-to-year standard deviation

Current practice is expensive before any rule is proposed. On development cases, computing the annual peak from recorded days alone — the default once a station-year passes a completeness screen — gives a mean absolute error of **1.305 °C, or 1.23 interannual standard deviations** of the quantity being measured, with 24.9% of station-years wrong by more than 1.0 °C. Substituting a day-of-year climatology over the gap gives 0.402 °C, the best supported reconstruction 0.225 °C with air forcing available for the whole panel (0.242 °C without it), and an oracle over the estimator set 0.130 °C. The error is therefore dominated not by which station-years are kept but by whether the record is repaired at all, which fixes the size of the question the rest of the paper addresses: given that a year has been repaired, which repaired years can be trusted.

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Table 3: Discrimination of a consequential error by axis (AUC), development cases, each endpoint at 0.47 of its own interannual standard deviation.

endpoint	peak-window overlap	warm-season coverage	outage length	AUC difference (95% CI)
hottest 7-day mean	0.823	0.824	0.752	+0.072 (0.063, 0.096)
hottest 30-day mean	0.819	0.820	0.750	+0.070 (0.061, 0.098)
amplitude	0.758	0.792	0.768	+0.024 (0.015, 0.037)
annual mean	0.672	0.818	0.845	−0.027 (−0.036, −0.018)
coldest 30-day mean	0.447	0.619	0.803	−0.184 (−0.228, −0.156)

4.1.2. The share of the support a gap hides orders every endpoint, and the ordering reverses

Table 3 reports, for five endpoints, how well each candidate property of the gap discriminates a consequential error. The ordering reverses between extremes and averages, every clustered interval excludes zero, and the winning axis is unchanged at tolerances of 0.25, 0.47 and 1.0 interannual standard deviations. The seven-day peak has the largest timing advantage of any endpoint, which the support account predicted before it was measured because a shorter extreme is more localised in time; peak-window overlap has an AUC of 0.447 for the cold extreme, because hiding the summer peak carries no information about the winter minimum. The last row is not evidence that cold extremes are a length-governed exception, and reading it that way would be a mistake: the two support axes in the table are defined on the warm season. Built on the endpoint's own season — a climatological coldest-30-day window and the observed share of the cold months — the support axis beats the length rule for the coldest 30-day mean on the same development cases (AUC 0.876 and 0.897 against 0.819; clustered error-rate gaps −0.0050 and −0.0059, intervals excluding zero), and adds to length in a grouped out-of-fold model (AUC 0.905 against 0.813 for length alone). What the table shows is that the support must be defined on the season the statistic lives in; it does not license a warm-season axis for a cold endpoint. That cold-endpoint analysis is exploratory: it uses the development panel and does not touch the sealed confirmation.

4.2. The response is conditional: where the gain holds and where it disappears

The advantage of conditioning on support is not general. It depends on how wide the estimand's support is, it survives only in part under the gap calendar these networks actually keep, and it differs across agencies and across how strictly their daily values are checked. Reporting only the pooled number would hide all three dependencies.

4.2.1. The transition is continuous, and a fixed narrow window turns over near a three-month support

Table 3 evaluates the question at two points and reads the gap between them as a dichotomy. Sweeping the hottest-k-day mean of Equation (1) from $k = 1$ to $k = 365$ evaluates it everywhere (Table 4, Figure 4a). The advantage of a fixed 30-day peak window — the axis the screen actually uses — is flat at about +0.075 AUC while the estimand's support is a month or less, falls to +0.050 at two months and to +0.002 at three, and is negative beyond that. The 90-day grid point is statistically indistinguishable from zero (95% interval −0.018 to +0.026, positive in 55% of network-clustered replicates), and the first clearly negative grid point is 182 days, so the crossing is only located between 60 and 182 days: a bootstrap of the whole curve puts its 95% interval at [77, 107] days and it moves between 85 and 102 days across the tolerance grid, while the opened reserved panel's interval ([104, 136]) overlaps it rather than displacing it. We therefore report a range and not a day count. The same sweep supports a second, different statement: a conditioning window dilated around the estimand's own support keeps its advantage over whole-year completeness out to about 359 days (95% interval 351 to 364, and it fails to cross at all in 7% of replicates), ending only at the trivial $k = 365$ coincidence, and the advantage is a hump that peaks near 60 days rather than decaying monotonically. The mid-range crossing is thus a property of holding the conditioning window narrow and fixed, not of the support idea itself. Nor is one window enough across the continuum: a single 60-to-90-day window is within 0.0004 AUC

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Table 4: The governing axis as a function of the support of the estimand, development cases.

support width k (days)	interannual SD (°C)	AUC, peak window	AUC, missing fraction	difference (95% CI)
1	1.18	0.825	0.751	+0.074 (0.053, 0.097)
3	1.17	0.824	0.746	+0.078 (0.055, 0.104)
7	1.16	0.823	0.746	+0.077 (0.053, 0.104)
15	1.12	0.821	0.743	+0.078 (0.053, 0.106)
30	1.08	0.819	0.744	+0.075 (0.049, 0.104)
60	1.02	0.815	0.765	+0.050 (0.028, 0.078)
90	0.94	0.792	0.790	+0.002 (−0.018, 0.026)
182	0.74	0.728	0.823	−0.095 (−0.111, −0.077)
365	0.55	0.672	0.845	−0.173 (−0.192, −0.157)

of each endpoint's own optimum up to a 90-day support, but costs 0.083 AUC at the annual-mean end. These continuum numbers are exploratory analyses developed after the reserved panel was opened; they motivate the screening axis but are not the sealed USGS confirmation reported with the data description. The two regimes of Table 3 are useful ends of that exploratory scale, not a proven universal dichotomy.

4.2.2. On the estimands the standing guidance was derived for, the guidance is right

The sine-fit parameters also connect to a physical reading that is worth keeping separate from the statistical one: air–water phase and amplitude carry information about groundwater and riparian influence (Hare et al., 2023), and a support-conditioned screen improves the error of those parameters without claiming to identify the process behind them.

Johnson et al. (2021) fit a sine wave by linear regression and take a mean, a phase and an amplitude; they do not compute a hottest 30-day mean. Reproducing that estimator on our panel puts the two accounts on the same footing for the first time (Table 5, Figure 4c). A completeness threshold is the correct axis for the fitted mean, at 0.845 against 0.672 for peak-window overlap, and is within 0.02 AUC of the best axis for the phase. The ordering is the same under the authors' own protocol, in which the sine is fitted to whatever days are present with no repair at all: peak overlap 0.667, 0.743 and 0.590 against whole-year missingness 0.902, 0.842 and 0.898 for the mean, amplitude and phase, at base rates of 32% to 34% consequential against 10% to 12% after repair.

Two reporting rules come out of this endpoint. Phase is only identifiable when the fitted amplitude stands above the noise, and the floor depends on the residual structure rather than on the sample size alone. The uniform-sampling formula $\sigma\sqrt{2/n}/A$ gives $A/\sigma \geq 4.5$ for a 95% interval no wider than the phase tolerance used here, and building the design matrix on each case's own observed days leaves that figure essentially unchanged (median required $A/\sigma = 3.73$), so the visibility mask is not what limits the endpoint. The residuals are what limit it: with the empirically estimated lag-one autocorrelation (median 0.89) the same design-based interval requires $A/\sigma = 16.2$, and a HAC interval 11.7; an independent block bootstrap of the sine-fit residuals lands between them at 3.2× the i.i.d. width. Every case the fixed rule would retain has a design-based half-width above the phase tolerance, so the earlier $A/\sigma \geq 4.5$ floor was too permissive under the unrepaired protocol and is replaced by the design-based requirement; the episode is a reminder that a signal-to-noise rule computed at uniform sampling does not transfer to a record with autocorrelated residuals (`p1_mechanism_boundary/`). What survives is the ranking result below, which does not depend on the floor. Second, the fitted mean and phase are the regime the standing guidance was derived for, and on them it is right.

The third declared prediction failed here. We expected that for the phase, whose influence is largest a quarter-year from the peak, two 61-day windows around the spring and autumn limbs would beat both peak overlap and gap length. They beat peak overlap decisively (+0.249 AUC, 95% CI 0.231 to 0.270) and they do not beat gap length (0.831 against 0.836). Weighting every day by the influence function of Equation (3) instead recovers the advantage: 0.847, which beats the whole-year missing fraction by +0.013 (0.009, 0.017) and peak overlap by +0.265 (0.249, 0.285).

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Table 5: Axis comparison on the sine-fit parameters (AUC), development cases, each at 0.47 of its own interannual standard deviation.

estimand	peak window	missing fraction	outage length	influence-weighted	best axis
fitted mean	0.672	0.845	0.845	0.845	outage length
amplitude	0.724	0.812	0.815	0.831	influence-weighted
phase	0.583	0.835	0.836	0.847	influence-weighted

Table 6: Decomposition by outage arm.

panel	arm	cases	base rate (%)	length rule (%)	screen (%)	contrast, points (95% CI)
development	empirical	14995	7.4%	2.52%	0.38%	−2.13 (−3.45, −1.26)
development	stratified	25709	19.3%	14.58%	6.17%	−8.39 (−9.25, −7.47)
reserved	empirical	4041	4.2%	0.74%	0.11%	−0.71 (−2.36, +0.05)
reserved	stratified	6779	20.8%	16.59%	7.78%	−8.68 (−10.46, −6.51)

4.2.3. The gain disappears under the gap calendar these networks keep, because long gaps begin in autumn

The frozen holdout test was run on the pooled case set, which mixes two arms whose base rates differ by an order of magnitude, and they must be reported apart (Table 6). The declared five-point effect is met on the pooled and stratified populations and *not* on the reserved empirical arm, where the interval includes zero. The holdout headline −7.13 points is therefore an effect on a mixture that includes a designed arm. The sealed USGS confirmation likewise finds a much smaller empirical-arm gain than the pooled contrast, so neither number should be quoted as next year's agency experience under natural outages alone.

Why the empirical arm is null is itself a result. Across the 6472 real interior gaps in this archive, gaps shorter than 30 days and gaps of 30 to 89 days are close to uniform in start month ($\chi^2 = 4.0$ on 11 df, $p = 0.97$ for the 30-to-89-day class). Gaps of 90 days or more are not: **72.3% of them begin between September and January and 17.9% between March and July** ($\chi^2 = 86.1$, $p = 9.5 \times 10^{-14}$; Figure 5a). The gaps long enough to destroy a summer peak are seasonal shutdowns, and they start after the warm season has been recorded, so little is at risk and little can be screened out: 4.2% of reserved empirical cases exceed the tolerance with no screening at all.

A counterfactual that holds gap lengths fixed and moves only start month is reported in Figure 5b as an exploratory amendment after correcting year-end truncation in the placement code (`results/wrr_submission_ready/e1_maintenance_recompute/`). Those percentage-point swings are not sealed confirmation and are not used as agency recommendations.

4.2.4. Agency and assurance strata on the withdrawn transport seal

Agency- and assurance-stratified contrasts from the first sealed transport score remain withdrawn: the historical quality-checked stratum that reported about −6.2 points was dominated by the NVE secondary panel and does not satisfy the pre-registered primary-panel gate (`amendments/f03_primary_qc_diagnostic/`). External confirmation is carried instead by the wave-2 multi-provider panel under repaired QA (Section 2.2). Measurement-contract sensitivity on the development and reserved panels remains informative as retrospective context: documented-daily-mean strata give about −5.7 and −6.6 points respectively (`amendments/provider_contract_sensitivity/`; `measurement_provenance_dictionary.csv`).

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Table 7: Which climatological window a screen should condition on, development cases.

estimand k (days)	best window (days)	AUC there	AUC on own support	gain	support wander (days)
1	90	0.862	0.760	+0.102	16.7
3	60	0.858	0.766	+0.092	16.3
7	90	0.855	0.778	+0.077	16.5
15	60	0.853	0.802	+0.051	15.3
30	90	0.852	0.827	+0.025	13.7
60	90	0.860	0.855	+0.005	9.4
90	90	0.856	0.856	0	7.4
182	182	0.851	0.851	0	6.9
365	365	0.844	0.844	0	—

4.3. The mechanism: hidden support and donor information control the error

Two measurable quantities organise the pattern: the share of the estimand's own support that a gap hides, which enters the error through the influence function of the statistic, and the synchronous anomaly information the surrounding stations carry. Both are visible at decision time, and neither is a restatement of gap length.

4.3.1. A support must be located before it can be conditioned on, so the useful window is wider than the statistic

Two predictions declared for this round were not met, and both failures have one cause. We predicted that the axis restricted to the estimand's own support would be at least as good as any hand-chosen axis at every k, and that its advantage would decay monotonically. Neither holds: the advantage is a hump peaking at a 60-day support, and for supports of 15 days or less the fixed 30-day peak window beats the estimand's own support, by 0.064 AUC at $k = 1$.

Table 7 gives the reason. The gain from conditioning on a window wider than the statistic tracks the year-to-year wander of the realised window's location almost exactly: it is +0.102 AUC at $k = 1$, where the climatological window is the worst estimate of where this year's window will fall and the wander is 16.7 days, and vanishes at $k \geq 90$, where the wander has fallen to 7.4 days and the support is wide enough to contain its own location error. A screening rule therefore does not condition on the support; it conditions on the support *dilated by the uncertainty in the support's location*, and a dilation of roughly two location standard deviations on each side is what the data choose.

One consequence is interpretive rather than deployed. AUC at the best conditioning window is between 0.85 and 0.86 for every estimand from the annual maximum to the half-year mean on this opened panel, so a 60-to-90-day window can help *discrimination* analyses. The screen scored below still conditions on a fixed 30-day climatological window; we do not treat the wider-window AUC advantage as a verified method innovation, because it was never wired into the frozen admission rule or shown to improve retention or false-admit risk at the operating point.

4.3.2. Position and length both matter, and the size of each effect depends on the reconstructor

Whether position or length matters more is conditional, and the controlled comparison says so. Holding the reconstructed window fully hidden and varying only the gap length, the completed-year error still rises for every reconstructor tested: by +0.15 °C for donor regression (95% interval 0.11 to 0.20), +0.25 for the day-of-year climatology (0.21 to 0.29) and +0.98 for linear interpolation (0.82 to 1.13). Holding the length fixed and varying only whether the peak window is hidden at all moves the error by more still, and again by an amount that depends on the reconstructor: +0.36 °C for donor regression (0.29 to 0.44) at a 60-day gap and +0.87 for the climatology (0.71 to 1.04). Adding the air-temperature-driven reconstructor to the same crossed design gives the same ordering — fully hiding the peak window costs +0.44, +0.76 and +0.79 °C at 30, 60 and 90 days, and lengthening the gap at a fully hidden window costs +0.35 °C ([0.26, 0.43]) — so the pattern is not

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an artefact of any one reconstructor. What the reconstruction does with the hidden days is equally important: the window that achieves the reconstructed maximum is a different window from the true one in 56 to 61% of cases for the climatological and interpolation methods and in 37% for donor regression, rising to 90 to 94% when the true window is fully hidden. Where that happens the error is larger by 0.28 to 0.70 °C, so a screen that condition on the climatological window is protecting against a shift that the reconstruction itself introduces. Whether length still matters once the window is fully hidden depends on the reconstructor: it does for linear interpolation (+1.12 °C between 30 and 90 days, [0.75, 1.45]) and for the air regression (+0.12, [0.07, 0.17]), it does under the station-year paired design for the two climatologies, and it does not for donor regression (+0.02, [−0.01, 0.05]), whose neighbours carry synchronous anomaly information that the other methods lack.

Both properties therefore act, their effects interact, and the reconstructor sets the scale; the fitted models agree with that reading, since once the support is in the model the length term adds little in this feature set (permutation importance −0.004 against 0.194 for peak-window overlap, Figure 6c) while the nested feature sets still rise from AUC 0.742 with length alone to 0.845 adding the support, 0.858 adding the observed-peak margin and 0.884 adding donor quality (Figure 6b).

How much of that increment survives the choice of training networks is a separate question, and the answer is small. Drawing twenty independent network subsets at each size and holding the evaluation panel fixed, adding the support features improves the 70%-retention error rate by 0.23 points at ten training networks (95% interval over draws 0.14 to 0.32) and by 0.14 points at seventy-five (0.07 to 0.20); the sign reverses in two to five of the twenty draws. The sample-efficiency reading of the earlier single-draw curve does not survive: support with ten networks is indistinguishable from the generic feature set with twenty-five (+0.20 points, [−0.06, +0.44]) and worse than generic with fifty or more (+0.61, [0.38, 0.84]), which puts its equivalent at about eighteen generic networks out of one hundred and ten. The published non-monotonicity at fifty networks was a draw artefact: that point lies outside the whole twenty-draw range at the same size. What the support features buy is therefore a small, consistent ranking increment, not a cheaper model.

On the shape of the average response, binning development cases on both properties simultaneously and measuring the growth of error along each with the other averaged out, the hottest 30-day mean grows 3.5× along the missing-days axis and 9.6× along the peak-overlap axis, while the annual mean grows 10.2× and 1.5× respectively: the two ratios swap by roughly an order of magnitude in each direction. That ordering is a property of these estimators on this panel, not a law about incomplete records.

Two qualifications belong with that. Length is not irrelevant: holding peak overlap at zero, the consequential-error rate still rises from 0.6% at 1 to 30 days to 8.7% at 121 to 200 days. And in a linear score that already contains warm-season coverage the length term takes a small negative weight, which is collinearity — the two correlate at −0.86 — and not evidence that longer gaps are safer. That correlation is also why a timing-versus-length labelling is wrong: warm-season coverage is a count of missing days restricted to a window, which is a length measure with a domain restriction, and it is the domain restriction and not the timing that does the work.

Donor information enters the residual through quality rather than quantity. Halving the donor set by training-period correlation costs +0.084 °C (95% interval 0.064 to 0.110) when the comparison is re-run with the donor count matched exactly between the two arms, against +0.082 (0.061 to 0.112) under the original construction in which the paired cases can differ by up to seven donors (34% of cases). Replacing donor anomalies with phase-shuffled surrogates while preserving each donor's seasonal cycle costs +0.234 °C (0.149 to 0.299), so donors supply synchronous anomaly information and not merely a seasonal shape.

4.3.3. The calibrated bound is honest enough to interpret

Everything that follows uses a predicted error bound to rank station-years, so the bound is checked before it is used. On the reserved networks the nominal 90% interval covers 90.9% of cases marginally, inside the pre-declared band of 0.87 to 0.93, and 86.3% to 94.9% within every pre-declared stratum (Figure 7a). Median widths adapt with the quantity they are meant to track, from 0.06 °C when the peak window is untouched

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Table 8: Reserved panel at 70% retention, across tolerances.

screen	> 0.25 °C	> 0.5 °C	> 1.0 °C	under by > 1 °C
outage-length rule	17.2%	9.7%	3.4%	2.0%
peak-overlap rule	9.2%	4.2%	1.2%	0.7%
calibrated screen	7.1%	2.5%	0.4%	0.3%
random	23.9%	14.4%	5.6%	3.7%

to 1.87 °C when most of it is hidden.

Those strata pool across networks, and conformal validity assumes an exchangeability that fails between river networks. Coverage inside individual reserved networks has mean 0.904 and median 0.913 but a minimum of 0.778, with five of 36 networks below 0.85. The pooled interval is an honest statement about a population of networks and is *not* a guarantee at a single new site; calibrating on the per-network conformal quantiles at the 0.9 level puts no network below 0.85 for a 1.5-fold widening. The screening results below are identical under every calibration to four decimal places, because the rank order the screen uses is a monotone transform of the bound, so a reader who distrusts the intervals need not distrust the frontier.

4.4. What it changes: retained error, archive composition, and the decision

A screening rule is judged by the archive it leaves behind, so we close on three consequences: the error among retained station-years at equal retention, which stations survive, and what the two admission rules do to the 2422 records in this archive that really have gaps and have never been decided.

4.4.1. Conditioning on the support cuts the error rate at equal retention

At 70% retention on development cases, the share of retained station-years with a peak error above 0.5 °C is 2.9% for the calibrated screen, 5.6% for a rule on peak overlap alone, 9.2% for the gap-length rule and 15.0% for random retention (Figure 8a). On the 36 reserved networks scored as a retrospective network holdout (mixed prior access; see the prior-access ledger), the screen gives 2.5% against 9.7% for the length rule (Figure 8b): a contrast of −7.13 points, 95% network-clustered interval −8.97 to −5.36, against the five-point effect declared for that freeze. Both declared conditions are met on that holdout, in every donor regime (−6.6, −6.7, −7.2, all excluding zero). The first sealed external panels remain withdrawn after audit; the independent wave-2 confirmation (Section 2.2) is a separate score and does not revise this retrospective holdout figure. The holdout result is not carried by the largest agencies alone: case-equal −7.01, provider-equal −6.29, network-equal −5.82, station-equal −5.54 points, with a median across individual reserved networks of −5.0, 83% of networks favouring the screen, and a worst network of +2.8.

Placed at the operating points the published guidance implies rather than at an arbitrary retention: at the seven-week unfilled limit (49 days, 49.2% retention on the reserved panel) the length rule leaves 5.2% and the screen 0.3%; at the twenty-week imputed limit (140 days, 85.1% retention) the length rule leaves 11.3% and the screen 6.9%. The second of those is more permissive than an earlier version of this analysis assumed — it admits 85% of cases rather than 74% — and the screen's advantage there is 4.4 points rather than 6.2.

Most of the benefit does not require the calibrated machinery (Table 9). Fifty-nine per cent is available from a logistic regression on three variables and 57% from peak-window overlap with no fitting at all; method selection contributes 0.2 points. Fitted on development networks and applied unchanged to the reserved panel, the deployable rule is

$$\text{risk} \;=\; \text{logistic}\big(-1.302 + 2.070\,x_1 - 2.412\,x_2 - 0.00187\,x_3\big), \tag{4}$$

where x_1 is the share of the station's climatological hottest-30 window the gap hides, x_2 the share of May-to-September days that remain observed, and x_3 the length in days of the longest consecutive gap. It gives 4.2% against the length rule's 9.6% at 70% retention, a contrast of −5.44 points (−6.85, −3.96), and cuts severe underestimation of the peak from 1.9% to 0.6%.

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Table 9: Component ablation, development cases at 70% retention, each variant re-run through the full out-of-fold pipeline.

screen	error rate	points vs the length rule
outage-length rule (incumbent)	9.24%	—
three-variable logistic regression	5.53%	−3.71
peak-overlap alone, no fitting at all	5.62%	−3.62
three-feature gradient boosting	4.89%	−4.35
geometry and timing only (8 features)	4.86%	−4.38
drop donor features	3.78%	−5.46
drop the observed-peak margin	3.13%	−6.11
full screen	2.93%	−6.31
full screen, screening only (no method choice)	3.11%	−6.13

4.4.2. A five-per-cent false-admit cap is met without reading the labels

The operating point quoted in the previous version of this analysis — the largest retention whose realised false-admit rate stayed under five per cent — was chosen by looking at the errors of the records being admitted. No agency can do that. The honest version fixes the protocol first: five folds by network, three fold seeds, a retention grid from 10 to 95 per cent, and, for each fold, a threshold chosen on the calibration networks alone as the largest grid point whose calibration false-admit rate plus a one-sided network-clustered 95% bound stays under five per cent. The held-out fold is then scored once (`p0_02_frozen_risk_thresholds/`; every fold found an admissible point, 180 of 180). On the 49-network external panel the calibrated certificate admits 76.7% of candidates at a realised 3.9 to 4.1% false-admit rate, the fitted risk score 75.5%, the transparent peak-window rule 71.8% and the length rule 48.9%. At the same risk target the fitted risk score admits 27.1 points more of the pool than the length rule ([23.9, 30.3]) and the certificate 27.9 points more; against the peak-window rule the fitted score leads by 2.7 to 4.2 points depending on the fold seed, one of which has an interval touching zero.

Two honest qualifications belong with those numbers. The point false-admit rate is under five per cent in all 36 pooled cells, but in six of them the interval's upper end exceeds it (worst 5.9%), so the cap is met as a point estimate rather than as a guarantee for every archive. And the multiplicative conformal calibration turns out to be a scale change rather than a ranking change: on the external panel the calibrated and uncalibrated certificate bounds admit an identical set of cases, so no claim is made here that calibration improves screening at fixed retention. Its role is the absolute-scale decision rule (admit when the bound is at most 0.5 °C), which is a different comparison. The label-chosen frontier of the earlier draft was optimistic by 1.5 to 3.2 points for these scores and 6.1 points for the length rule — the gap between describing a frontier and deploying one.

4.4.3. The incumbent is not handicapped, including by the absence of the field's own reconstruction

A screening comparison can be rigged by pairing the incumbent with a weak reconstruction, so the incumbent is paired here with every reconstruction in the study, including one nobody can deploy (Table 10).

The screen beats the incumbent armed with a reconstruction nobody can run, and the reconstruction paired with the incumbent throughout is the strongest of the deployable options rather than the weakest. Air-driven reconstruction is reported twice, because the earlier label was wrong: air-temperature regression remains a workhorse for data-sparse basins (Sanikhani et al., 2026; You et al., 2026), and the library's air ridge with a fixed penalty ($\alpha = 1.0$, seven lags, whole-year fit, penalised) was described as "in the manner of Beaufort et al. (2022)" but shares only its input with that paper, and is now labelled as what it is. Reimplementing the cited method faithfully — per-station OLS of summer water temperature on a right-aligned moving average of air temperature, 2–10 day windows chosen per station by R^2, fitted on July–August — gives 12.76% against the library ridge's 15.16% (paired difference −2.40 points, [−3.20, −1.62], 1200 network-clustered draws), and both remain worse than per-case selection at 9.00% (`amendments/faithful_baseline/`). The axis conclusion does not depend on the estimator library: the timing-minus-length AUC contrast is +0.070 with the old forcing and +0.081 with the new. The screen uses the new information, selecting the air-driven

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Table 10: Development cases at 70% retention, with air forcing available throughout.

screen	reconstruction	error rate above 0.5 °C
length rule	information-conditioned policy (used throughout)	9.00%
length rule	always library air ridge, $\alpha = 1.0$, else climatology	15.16%
length rule	always faithful Beaufort et al. (2022) climate correction	12.76%
length rule	always donor ridge, else climatology	9.57%
length rule	always smooth climatology	19.79%
length rule	oracle over methods, not deployable	3.62%
calibrated screen	deployable estimators it selects itself	2.87%

Table 11: Archive composition under each admission rule, on the 2422 real incomplete station-years.

rule	admitted (n)	admitted years with the climatological peak window hidden (n)		
		any day	more than half	wholly
repair everything, no screening	2422	696	261	186
length $\leq$ 140 d, with imputation	2221	507	82	17
length $\leq$ 49 d, unfilled	2016	437	45	2
peak window $\leq$ 10% hidden	1967	241	0	0

estimator for 20.0% of cases against 0.07% before, and donor regression for 53.1%.

One result here was not expected: the *fixed* information-conditioned policy gets slightly worse at donor-less stations when air becomes available, from 0.340 to 0.364 °C.

4.4.4. On records that really have gaps, the two rules disagree on one year in six

Every number above is computed on a complete station-year with a synthetic hole punched in it, which is the only way to have a truth. The same archive contains 2422 genuinely incomplete station-years, a third of the total, which carry no truth and have never been decided. Three things can be asked of them without inventing one (Figure 9).

They are not a marginal population. The median incomplete year is missing 9 days and the 90th percentile 128; 28.7% of them have at least one day of the station's climatological peak window missing, and 261 have more than half of it missing.

The two rules disagree on a sixth of them. At the seven-week unfilled limit the length rule admits 83.2% and a peak-window rule at 10% hidden admits 81.2% — near-equal retention, which is the comparison this paper insists on — and they **disagree on 15.1%** of station-years: 207 are admitted by the incumbent and rejected on the support, and 158 are rejected by the incumbent and admitted on it. At the twenty-week imputed limit the incumbent admits 91.7% and the disagreement is 11.6%, of which 267 are in the dangerous direction and 13 in the wasteful one. The disagreement is present in every provider with more than 20 incomplete station-years, from 11.1% to 25.7%.

What the disagreement costs can be counted without a truth (Table 11). The published guideline agencies are most likely to apply, the twenty-week limit that exists because imputation is assumed, admits 17 station-years whose *climatological* peak window was never observed at all and 82 whose peak window is mostly gone. That count is a proxy and its error rate is measurable: on the sealed designed cases, where the realised peak is known, a wholly hidden climatological window carries a positive predictive value of 0.74 (network-clustered interval 0.71 to 0.76) for a wholly hidden *realised* peak, with sensitivity 0.87 and only 3.1% of proxy-clear cases hiding the realised peak. Transported to the 17 flagged station-years, that implies about 12 to 13 whose realised annual peak was wholly unobserved and about 4 whose proxy fired while the realised peak was visible. A risk model fitted on development cases from decision-time features only, and checked for calibration on the reserved panel first (largest decile miscalibration 6.2 points, and under 1.6 points in nine of ten deciles), predicts a consequential-error rate of 0.98% among the years the unfilled limit admits, 1.85% among those the imputed limit admits, and 0.26% among those the peak-window rule admits at essentially the same retention. That last figure is an extrapolation from scored cases to unscored records and is labelled as one; Section 4.4.9 tests the exchangeability it assumes.

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At the level of a regional aggregate the choice of rule barely matters: admitting incomplete years at all raises the archive's mean annual peak by about half a degree, and the four rules differ by less than 0.1 °C (+0.52, +0.51, +0.53 and +0.45) and by 0.4 points in the share above 20 °C (+4.55 to +4.17).

The same is true of a multi-year slope. Theil–Sen slopes of the annual peak have RMSE 0.143 °C yr^{-1} on recorded days, 0.044 with reconstruction and no screening, 0.048 under length screening and 0.041 under the screen: the dominant step is again whether the record is repaired. Eleven complete years per station is in any case too short to resolve a decadal trend, so we report the slope error as a diagnostic of the reconstruction and make no trend claim.

Per station-year the choice does matter, and Section 5.3 follows it into the judgement an agency actually makes.

4.4.5. The hot-year composition of the admitted set, decomposed

An agency that screens also changes the mix of years it keeps, and the honest way to report that is a decomposition rather than a single number. Writing the predicted hot-year share among admitted records as $\hat{p}_A$, the true share among them as p_A and the true share in the full candidate pool as p_{full}, the total deviation splits exactly into a prediction term and a selection term, $\hat{p}_A - p_{\text{full}} = (\hat{p}_A - p_A) + (p_A - p_{\text{full}})$, and both are reported for every rule (`p0_04_env_closed_loop/`). On the reserved panel at 18 °C the prediction term stays below half a point for every rule (largest magnitude 0.46 points, for judging everything), while the selection term reaches 1.40 points for the peak-window rule and 0.78 for the length rule; for the calibrated screen it is 0.36 points, and the total deviation is 0.29 points. The repair matters less for the magnitudes than for the attribution: judging every record without screening previously reported a deviation of exactly zero even though it made 172 wrong judgements, and its corrected deviation is its own prediction error. Two further cautions follow from the same table. The elevation-balanced variant's apparent advantage in hot-year composition has the opposite sign once records are weighted by network rather than by case, so no cost-free balancing claim is made. And the length rule's own counts depend on how ties at the retention cut are broken, moving its errors by five in the reserved panel across thirteen seeds, whereas the calibrated and abstaining rules are unchanged across all of them.

Abstention is compared against judging everything at the same error budget rather than only against the screen. At a budget of 0.5% wrong judgements, judging everything is admissible in only one of four panel-threshold cells (the external panel at 18 °C, where it makes 52 wrong judgements in 13 017 candidates, 0.40%); there abstention can at best tie it. In the three cells where the budget binds, interval abstention holds the largest coverage inside the budget (96.1% at 0.48% on the reserved panel at 18 °C, 94.8% at 20 °C, and 96.7% on the external panel at 20 °C). At equal coverage, abstention usually has fewer errors than the calibrated screen (none against 25 at 70% coverage on the reserved panel). Abstention therefore wins under one criterion in the external 18 °C cell and under the other elsewhere, which is the comparison the earlier draft skipped.

4.4.6. The final rule runs on real calendars and knows when it cannot

The rules compared on the real gap calendar were, until this revision, the length rule and the peak-window rule — not the certificate rule the paper recommends. Re-running the replay with the final rule, its threshold calibrated on network-disjoint data, on 418 candidate groups from 14 networks (22 cross-year placements evaluated in both affected calendar years, with the declared gap length asserted against the mask in every case), the rule neither loses to the simple rules when neighbours fail nor buys its position by abstaining. With every donor masked it decides 94.5% of candidates, against 94.3% for the peak-window rule and 77.8% for the length rule; among the cases it decides its consequential-error rate is half a point better than the peak-window rule (−0.51 points, [−1.19, 0.00]) and half a point worse than the length rule (+0.51, [0.00, 2.17]), whose stricter threshold admits a fifth fewer records. The share of records that are both produced and within 0.5 °C exceeds the length rule by 16.3 points ([9.8, 24.3]), while against the peak-window rule it is a tie (+0.7, [0.0, 1.4]). Realised error among its decided cases is 1.0, 0.5 and 0.5 per cent across the three co-failure levels, well inside the five per cent its calibration promised, so the threshold transfers conservatively from designed

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Table 12: Error against representativeness, reserved panel at 70% retention.

rule	error rate (%)	ρ with elevation	donor spread (points)	retention SD (points)
outage-length rule	8.81%	−0.10	4.5 pts	0.075
peak-overlap rule	3.75%	+0.16	2.4 pts	0.104
calibrated screen, global threshold	2.53%	−0.27	4.3 pts	0.149
screen, balanced within donor regime	2.52%	−0.27	0.1 pts	0.148
screen, balanced within elevation tertile	2.85%	+0.08	4.6 pts	0.148
screen, balanced within both	2.84%	+0.07	0.1 pts	0.144

outages to real calendars.

The accounting that makes those numbers interpretable is the one the earlier replay did not keep. When every donor is masked the donor reconstructor cannot be formed at all, yet the previous code returned a finite donor value for 412 of the 418 cases — the hottest fully observed window, with a mean absolute error of 0.24 °C against 0.04 °C for real donor reconstruction. The donor estimate is now unavailable in that situation, the strict rule does not admit the case, and a separately labelled fallback variant reports it under the observed-days estimator it actually is; the three paths are never merged into one robustness column (`p0_03_final_method_replay/`). The replay transplants real calendars onto other years of the same station, so it is a within-development cross-fitted probe rather than a sealed confirmation, and the co-failure levels are imposed by masking donors with the target calendar rather than observed.

4.4.7. Conditioning on difficulty removes mountain catchments unless the threshold is applied within strata

A rule that lowers the average error by discarding hard station-years has bought accuracy with representativeness, and in this application the hard sites are not a random subset: Section 4.4.8 showed they are high-elevation, steep, donor-poor catchments, which are the cold-water headwaters the summer endpoint exists to describe.

At 70% retention the calibrated screen's per-station retention correlates with site elevation at Spearman $\rho = -0.26$ ($p = 0.011$) on development and -0.27 ($p = 0.077$) on the reserved panel, while the incumbent length rule is elevation-neutral at $\rho = -0.05$ ($p = 0.60$) and -0.10 ($p = 0.52$). On the reserved panel the screen retains 77.7% of station-years in the lowest elevation tertile and 66.9% in the highest, a spread of 10.8 points against 1.7 for the length rule (Figure 10b). By donor regime the same pattern appears on development cases: 57.1% at donor-less stations against 72.5% at donor-rich ones, a 15.4-point spread, while the length rule tilts the other way.

The distortion is cheap to remove (Table 12). Applying the screen's threshold within stratum rather than globally costs 0.31 points of consequential-error rate on development cases and 0.30 on the reserved panel and flattens both distortions. Stratifying on donor regime alone does not fix the elevation tilt and stratifying on elevation alone does not fix the donor tilt, so an agency that cares about both must declare both.

4.4.8. Where it is hard: catchment attributes explain a modest share of the error

The account so far is a property of estimators and estimands and would hold for any time series. For the 140 United States stations matched to GAGES-II across 57 networks, we regressed donor synchrony and completed-year error on basin attributes with the expected direction of each effect declared before the regressions were run (Supporting Information Figure S7). Site elevation and basin slope both reduce donor synchrony (Spearman $\rho = -0.29$, 95% CI −0.46 to −0.08, and $\rho = -0.31$, −0.51 to −0.07), and site elevation raises both completed-year error ($\rho = +0.24$, 0.06 to 0.43) and the consequential-error rate ($\rho = +0.25$, 0.08 to 0.43).

The groundwater prediction taken from the literature is not reproduced, and the reason is instructive — with the caveat that the comparison is indicative rather than a replication, because the original claim concerns

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imputation harm to specific fitted sine parameters while the contrast here is the completed-year error of a different estimator. Johnson et al. (2021) report that filling can harm groundwater-influenced sites, which predicts larger completed-year error at a high baseflow index. Baseflow index does reduce donor synchrony in the predicted direction, from 0.70 in the lowest tertile to 0.59 in the highest, although the rank correlation interval includes zero ($\rho = -0.19$, -0.42 to 0.10). But the error moves the other way: splitting at the median baseflow index, high-baseflow stations are -0.043 °C easier than low-baseflow stations (95% CI -0.081 to -0.004). Across baseflow tertiles the mean annual peak falls from 22.7 °C to 18.1 °C ($\rho = -0.47$, -0.60 to -0.26) and its interannual standard deviation from 0.98 to 0.76 °C (Supporting Information Figure S7c), so two effects act in opposite directions and the second is larger.

Regulation does not make station-years harder for this endpoint: reference and non-reference basins differ by +0.004 °C (95% CI -0.051 to +0.093), although the reference flag is a coarse proxy for regulation rather than a controlled contrast, so this bounds nothing about individual dams. Catchment attributes explain a modest share of the variation, a standardised linear model reaching $R^2 = 0.22$ for donor synchrony and 0.13 for completed-year error, and no attribute predicts how much a screen gains: every correlation between a basin attribute and the per-station screening benefit has an interval containing zero. Whether an attribute predicts what a screen *keeps* is a different question, and Section 4.4.7 answers it differently.

4.4.9. The conclusion survives instrument error, a stricter fitting protocol, and transfer across agencies

A rule that is fitted on earlier years and applied to later ones is the design an agency actually faces, so it is run separately from the leave-one-year-out protocol. Splitting each station's complete years into an earlier 60% and a later 40% block, fitting and calibrating on the earlier block alone and evaluating once on the later one, the ranking conclusion is unchanged: the contrast at matched 70% retention is -6.8 points ($[-7.7, -5.7]$, 110 networks), and the difference from the leave-one-year-out protocol is -0.07 points ($[-0.33, +0.24]$). What the past-to-future design does change is what can be promised at a risk target: with the threshold chosen on the earlier block only, the risk score keeps 81.2% of the later block against 41.2% for the length rule, a retention gain of 40.1 points ($[28.0, 49.1]$) — but the realised false-admit rate on the later block is 5.9% against the 5% the calibration block promised, so the honest reading is a retention gain at a calibrated target whose level is not guaranteed out of block, not a guaranteed risk cap on future years. Warm years are not a failure mode: the contrast is -7.7 points among hot years ($[-8.7, -6.6]$) and -5.3 among ordinary years ($[-6.5, -4.3]$).

The same rule is then re-run inside each measurement-contract stratum. It keeps its sign everywhere but not its size: -5.6 points on the 84 networks whose providers document a daily mean ($[-6.4, -4.9]$), -8.4 on the 14 networks whose sub-daily values the project aggregates ($[-10.3, -5.0]$), -4.9 on the ten networks whose provider never states the statistic ($[-7.6, -2.5]$), and -11.4 on the two Finnish networks whose documented value is a morning reading, where two networks are too few for an interval. So a sentence about transfer across agencies should quote the documented-daily-mean figure, -5.6 points, rather than the pooled one. Since the stratum is a property of the archive rather than of the method, an agency adopting the rule should expect the documented-daily-mean figure for its own records and not the pooled one.

Missingness is weakly informative, and the effect is bounded well below the tolerance. Every case transplants a mask onto a year whose values were observed, which is valid only if the mask is ignorable in the sense of Rubin (1976); loggers are lost to floods, buried by ice and fouled in low flow, and each of those coincides with anomalous conditions. Across the 6472 real gaps, the mean anomaly of the seven days preceding a gap, relative to the station's own day-of-year climatology, is +0.018 °C for gaps of a week or less ($p = 0.44$), +0.038 for 8 to 29 days ($p = 0.56$), -0.002 for 30 to 89 days ($p = 0.98$) and -0.215 for 90 days and longer ($p = 0.081$). Donors recording through a gap run +0.063 °C warm during short gaps, detectable at this sample size ($p = 0.017$) and 3% of the within-station anomaly scale. The design is therefore defended on the magnitude of the effect rather than on its absence, and the class that warrants a limitation is gaps of three months and longer. What the records support is a specified-shift sensitivity, not a bound: on the day-level recompute, where the reconstructed hidden days are offset by δ °C and the hottest 30-day window is re-sought, the screen keeps its advantage at equal retention for δ from -1 to +1 °C (-10.7 points at $\delta = -1$,

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network-clustered interval −13.7 to −5.4), and 28.8% of cases switch to a different window once the offset is applied (`amendments/mnar_day_level/`: 240 cases, six networks, an exploratory sensitivity on a reserved subset whose own status is *not an identification bound*). Because the perturbation is a constant offset on hidden days rather than a model of why they are missing, this establishes what the screen tolerates, not what the missing values were; the approximate fixed-window arithmetic used in an earlier version is withdrawn.

Three further checks move the numbers and not the conclusion. Instrument error in these archives is systematic and time-varying as well as random (Jackson et al., 2025), which is why the check below is repeated under correlated, drifting and step-like error and not only under white noise. Perturbing the reference series by 0.15 °C of normal noise moves the development contrast from 5.98 to 6.03 points, the reserved contrast from 6.21 to 6.22 points, and the mean absolute error from 0.2361 to 0.2363 °C; the paired change is +0.049 points (−0.036, +0.150) on development and +0.013 (−0.175, +0.185) on the reserved panel, so the check no longer reports a reduction in the advantage (an earlier version evaluated the two arms with two different reconstruction choices). Correlated error displaces the 30-day endpoint harder at the same daily scale and still leaves the contrast within its interval: an AR(1) process with $\rho = 0.7$ by 2.25 times and a mid-year 0.3 °C step by 2.12 times, giving 5.95 and 6.21 points (development) and 6.19 and 6.43 (reserved). A constant annual bias of ±0.5 °C is an exact endpoint shift, so it moves the contrast by at most 1.0 point (largest change interval 0.35 to 1.82 points, on the reserved panel at −0.5 °C) while raising the mean absolute error to 0.30 °C. Restricting estimators to years before the gap raises reconstruction error from 0.242 to 0.270 °C on a panel that falls to 33633 development cases, and moves the contrast from −6.35 (−7.12, −5.52) to −5.92 points (−6.72, −4.99) with coverage 0.906. Under leave-one-provider-out the contrast is negative with an interval excluding zero for all nine agencies, from −11.6 points (Finland) to −1.2 (Switzerland), with conformal coverage between 0.82 and 0.96.

5. Discussion

Using 5114 complete river station-years, 51524 designed outages and two sealed panels, we have shown that the error of an annual thermal extreme is governed by where a gap falls rather than by how long it is, that an annual mean reverses that ordering, and that screening on the hidden share of the statistic's support leaves a materially better archive than a completeness threshold at equal retention. The result agrees with the standing guidance on the estimands that guidance was derived for, and disagrees with its transfer to extremes; the disagreement is one of conditioning axis, not of arithmetic. Two mechanisms account for it, one of which the data can measure and one of which they cannot.

5.1. Two mechanisms explain the reversal, and one of them is not in the data

The transition measured in Table 4 is not a property of rivers, of the estimators, or of the screening rule. It is a property of the functional being computed, and it follows from where that functional puts its weight. A statistic whose influence is concentrated on a few days is destroyed by losing those days and is almost unaffected by losing any others; a statistic whose influence is spread evenly over the year tends to degrade with how many days go missing and is less sensitive to which. Between those two limits the influence is neither concentrated nor uniform, and the discriminating power of a narrow conditioning window falls as the weight it captures falls. On this panel's exploratory width sweep, a fixed 30-day window stops beating a whole-year count near a three-month support; that numerical crossing is panel- and climate-dependent, not a universal constant.

Two measurements distinguish that account from a re-description of the data. First, the gradient ratios swap by roughly an order of magnitude in each direction between the peak and the annual mean, which is a direct measurement of how error grows along each axis rather than an inference from discrimination. Second, and more decisively, the account makes a prediction about a case it was not built on. The sine-fit phase has an influence function with two broad lobes a quarter-year from the peak, so the account predicts that peak-window overlap should be a poor axis for it — and it is, at 0.583 against 0.836 for gap length. A description

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fitted to the extreme-versus-average contrast would not have predicted that.

The same account explains where it failed. We predicted that conditioning on the estimand's own support would always be best, and it is not, because the support has to be estimated and the estimate wanders. Once the wander is measured the failure becomes a quantitative statement rather than an exception: the gain from widening the conditioning window tracks the wander from 16.7 days at $k = 1$ to 7.4 days at $k = 90$ and disappears when the support becomes wide enough to contain it. This is the part of the result we would expect to transfer to other variables and other climates, because nothing in it is specific to water temperature: any estimator conditioned on an estimated support inherits the uncertainty of that support.

5.2. Where this agrees with the standing guidance, and where the disagreement lies

An earlier framing of this work held that the incumbent rule gets an ordering backwards. Section 4.2.2 shows that reading is wrong on the guidance's own terms. Johnson et al. (2021) derive their limits for sine-fit mean, phase and amplitude, and for those three estimands a completeness threshold is the correct axis or within 0.02 AUC of it. What we correct is not the threshold but the habit of carrying it across to an estimand it was not derived for.

That reading also resolves the anomaly the guidance itself reported. Its authors observed that the timing of consecutive missing data had "inconsistent effects ... among regions, years, and individual parameters" on annual thermal-signal estimates. Their three estimands have three different influence functions — a constant, two lobes at the extremes, two lobes at the transitions — and only one of the three has weight concentrated narrowly enough for a timing effect to appear. Observing all three at once, inconsistency is the only thing one could see. We do not contradict the observation; we decompose it, and we can now say in advance which of their parameters should show a timing effect and which should not.

A smaller result points the same way and was not expected. The fixed information-conditioned policy gets slightly *worse* at donor-less stations once air forcing becomes available, because its preference order was written when air was rare and it now prefers an air regression to a climatology at sites where the climatology is better. A fixed ladder over estimators is not robust to a change in what is available; a per-case selection is. That is an argument for the method of this paper rather than against it, and we report it because it was the opposite of what we expected.

The disagreement with the gap-filling literature is of a different kind and is not resolved by our result. Beaufort et al. (2022) fill every record alike, and on our panel pairing the incumbent rule with that treatment makes it worse rather than better (Table 10). That is not evidence against air-temperature regression, which is the best available estimator at many of our stations and which the screen selects for 20.0% of cases. It is evidence that a reconstruction chosen once for a whole archive and a reconstruction chosen per station-year are different objects, and that the quantity a screening comparison must hold fixed is the second.

5.3. What changes in the judgement an agency actually makes

The archive-level aggregate is the wrong place to look for the value of a screening rule, because it averages over the station-years where the call is easy. A criterion-exceedance judgement is the right place, because it is made one station-year at a time and it is where a wrong peak becomes a wrong decision. On reserved networks the misclassification rate of a 20 °C exceedance is 6.86% when the statistic is computed on recorded days, 1.97% when every year is reconstructed and not screened, 1.53% under length screening and 0.36% under the screen. Restricted to the years where the call is close the four figures are 16.84%, 9.50%, 7.26% and 1.92% within 1.0 °C of the threshold and 19.10%, 13.12%, 10.24% and 3.38% within 0.5 °C, so the advantage is proportionally larger exactly where it is needed. The bias in the count of years above 20 °C falls from −0.76 years per station under recorded days and −0.32 under length screening to +0.06.

That result is not an artefact of the threshold chosen. Swept from 14 to 24 °C the screen's misclassification stays between 0.18% and 0.37% while the length rule stays between 0.84% and 1.53%. The sweep matters because no single number is the criterion: regulatory practice uses a seven-day average of daily maxima

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with criteria differing by several degrees between jurisdictions and designated uses, 16 °C for core cold-water salmonid habitat and 13 °C for salmonid spawning (Washington State Department of Ecology, 2024), which is why the seven-day endpoint is reported throughout and why the threshold is swept rather than asserted. The illustration is not a compliance test, and we can now say by how much it misses one: on the 1591 station-years of the sealed USGS panel that publish both statistics, the regulatory seven-day average of daily maxima sits a median +1.57 °C above our seven-day average of daily means (5th to 95th percentile +0.49 to +4.43), and at criteria of 16, 18 and 20 °C the two endpoints disagree on 5.5%, 4.3% and 10.4% of station-years — every disagreement a missed exceedance and none a false one (`amendments/regulatory_daily_max/`). A threshold shift matches counts but not decisions, because the offset is nearly constant (slope −0.002 per °C). An agency applying a daily-maximum criterion must compute it from daily maxima, which the archive publishes for 90 of the 91 sealed stations.

For a judgement of this kind, screening is not the best use of the calibrated object. Discarding a station-year is a blunt response to uncertainty about a single call. On the reserved retrospective panel, absolute admission at a 0.5 °C calibrated bound retains 67.5% of years with a 2.0% consequential false-admit rate (uncalibrated 66.2% / 1.8%); the multiplicative scale barely moves ranking-based risk-cap retention, which is why calibration must be judged on absolute admit sets, not on risk–retention frontiers alone. For threshold calls, abstaining only where the conformal metric interval straddles 18 °C judges 93.1% of reserved years at 0.19% misclassification among judged calls, versus judging 70% under calibrated screening at 0.33% — and combining screen with abstention judges 68.3% at 0.03%. The two actions answer different questions — screening decides what enters an archive, abstention decides what one judgement may say — and the same calibrated object serves both.

5.4. What an agency should change, and in what order

The first recommendation is to treat screening as estimand-specific. An agency assembling warm-season extremes from maximum gap length should trial conditioning on how much of a station's climatological warm window the gap hides, and should keep length-based rules for annual means and the sine-fit regime parameters for which standing guidance was derived. Cold extremes are not an exception to this: the support axis has to be defined on the cold season, and when it is, it beats the length rule for the coldest 30-day mean on the development panel; a warm-season support axis is the wrong tool there, which is what the reversal in Table 3 shows. On the retrospective reserved holdout, substituting a support-aware screen takes the consequential-error rate from 9.7% to 2.5% at equal retention on pooled designed outages; the empirical-arm gain is not resolved, so this is not yet a claim about the networks' natural failure calendar.

Maintenance-calendar pricing that previously appeared to show autumn servicing gains of several percentage points is withdrawn from recommendations until the cross-year outage placement is recomputed: the earlier counterfactual truncated late-year gaps at day 366 while still labelling them by their nominal length. Until that recompute is confirmed on networks outside the development programme, schedule changes remain a hypothesis suggested by autumn-heavy long outages, not a verified error-rate benefit. It cannot be replaced by an observed counterfactual on this archive, and we checked rather than assumed that: for the 200 USGS stations of the incomplete-year population we classified all 2921 real interior gaps and 68180 missing days against the provider's own daily-value qualifiers, and no cause-naming code — maintenance, equipment, ice, seasonal or discontinued — occurs once. What the qualifiers do support is an archival decomposition: 94.9% of missing days are days the provider published no record for at all (interval 88.7 to 98.2), 4.8% were published only under a different statistic of the same parameter, and 0.3% were published for the target statistic and dropped by the archive (`amendments/outage_causes/`). A rule that excludes planned-maintenance gaps and keeps equipment failures is therefore a no-op on 0 of 1361 station-years, while re-ingestion would recover five station-years (all 2024–2025) whose target values the live service now serves. Redundancy investments similarly need cost and failure-mode evidence beyond synchrony correlations on the opened panel.

Thresholds should be applied within strata when difficulty correlates with elevation or donor poverty, or the archive can quietly lose mountain stations. Nothing here applies to ungauged sites, because locating support uses a station's own history. A pooled conformal interval is a statement about a population of networks, not

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an automatic guarantee at a single new site; cluster-aware calibration must be chosen on calibration data only and validated on networks outside the development programme before it is sold as site-level coverage.

5.5. What generalises, and what does not

How much evidence a confirmation would need is a design question that can be answered from the panels in hand. The contrast between the screen and the length rule has a small intra-network correlation but networks hold hundreds of cases each, so the design effect is large: the current development panel behaves like 14.1 independent networks and the external panel like 11.0. Simulating the study by resampling whole networks, twenty networks reach 53% power for a two-point effect and full power for the five-point effect the freeze declared; eighty networks reach 98% for two points, with a minimum detectable effect at 80% power of 1.33 points (`p0_05_power_independence/`). The same simulation exposes a caveat that should travel with any power statement here: at twenty networks the interval excludes zero in 18% of draws when the true effect is exactly zero, so the clustered interval under-covers at small network counts and the detectable-effect figures should be read with that in mind. Network counts, not case counts, are the currency of this design.

Three parts of the result are portable. The governing quantity of Equation (3) is defined wherever a derived statistic is computed from an incomplete record, and its ingredients — an influence function and a mask — are available in any such setting. The protocol that measures the transition is portable in the same way: compute the statistic over a family of support widths, measure how far the support's location moves between periods, dilate by that amount, and condition on the result. And the finding that a screening rule must be specified together with its strata is a statement about selection, not about temperature.

Three parts are not. The numerical crossover near three months depends on how concentrated a river's thermal year is and on how far its peak wanders between years; both are properties of this climate and this variable. The operational gain is regime-dependent: designed-outage contrasts on the retrospective holdout are larger than empirical-arm contrasts on the same networks (about −0.7 to −1.9 points), so an agency must not quote the designed mixture as a bankable operational guarantee. The withdrawn first seals are not re-used; the wave-2 confirmation ($\Delta = -5.9$ points) is the external score that enters conclusions. Catchment associations rest on 140 of 204 stations in one country and should be read as attribute correlations on that subset, not as a geology-versus-terrain law.

One boundary deserves naming separately, because it limits interpretation rather than transfer. The mechanism argument is supported by an influence-function derivation, by measured error gradients, by a continuum with a located crossover and by an out-of-sample prediction about the phase. It is not supported by an independent observation of the internal quantity itself, because the hidden mass of Equation (3) is by construction unobservable on the records it matters for. A different representation that reproduced the same gradients would not be excluded by our evidence.

5.6. What the remaining uncertainty changes, and what it does not

Uncertainty enters this study in four places, and it is worth separating what each one moves. Measurement error in the reference series changes the numbers in the third decimal and changes neither the mechanism nor the recommendation: 0.15 °C of noise moves the reserved contrast from 6.21 to 6.22 points, and time-correlated error displaces the endpoint twice as hard while still leaving the contrast inside its interval. Fitting protocol changes the numbers by about half a point and changes nothing else: restricting estimators to prior years moves the contrast from −6.35 to −5.92. Neither of these is capable of overturning a conclusion.

The other two are. The choice of population changes the practical conclusion completely: on the arm built from the gap geometry these networks actually experience the effect is −0.71 points with an interval containing zero, and any reader who needs a number for their own archive needs the arm that matches their gap regime rather than the pooled figure. And the absence of an independent observation of the internal quantity limits the mechanism judgement, as the previous section states: we can show that conditioning on an estimate of the support discriminates better, and we cannot show that no other representation would do

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as well.

Two further limits belong here because they are specific rather than generic. Missingness is weakly informative and the transplant design is defended on the magnitude of that effect rather than on its absence, which makes the defence weakest for gaps of 90 days and longer, where the leading flank runs 0.215 °C cold. And the archive-level consequence is small: the choice of rule moves a regional mean annual peak by less than 0.1 °C. We report that because it deflates a claim we might have made. At the level of a regional aggregate the decision that matters is whether to repair; the rule matters per station-year, and it matters for who is in the archive.

6. Conclusions

Using 5114 complete station-years from 146 river networks and 51524 designed outages, with a retrospective network holdout and an independent wave-2 confirmation panel, we quantified what an incomplete year can still support. Three answers follow.

First, indicator-conditioned risk ranking is worth adopting for hottest-30 admission under a stated tolerance: the frozen screen cuts consequential errors on the retrospective holdout from 9.7% to 2.5% at 70% retention (−7.1 points, [−9.0, −5.4]), and on the wave-2 designed confirmation by −5.9 points ([−6.9, −4.9]). The same rule can be deployed under a risk cap rather than a retention target: with the threshold chosen on calibration networks only, it admits 76.7% of external-panel candidates at a realised false-admit rate below five per cent, against 48.9% for the gap-length rule, a retention difference of 27.9 points ([24.8, 30.4]) at the same risk. That comparison is the one an agency can reproduce, because no evaluation label enters the choice. Peak-window overlap remains a strong transparent rule and explanation, but on the development panel a same-capacity generic risk model already matches a model that adds engineered support share; the claim is admission by fitted indicator risk, not that support share is an irreplaceable feature. Fixed-reconstructor crossed controls show that fully hiding the peak window raises error under climatology and donor methods alike, with network intervals excluding zero, while length effects at fixed overlap are weaker and method-dependent — conditional mechanism evidence, not a universal geometry law. The withdrawn first seals and the failed transport quality-checked gate remain labelled and are not re-used.

Second, the gain is conditional in ways a pooled number hides: it is larger on stratified pressure outages than on empirical transplants (wave-2 empirical about −0.8 points; reserved empirical interval includes zero), and it depends on the estimand. Completeness thresholds remain the correct axis for annual means and the sine-fit parameters, which is where the standing guidance was derived. Wider conditioning windows that raise AUC are an applicability boundary for discrimination, not a deployed component of the scored screen.

Third, the practical consequence is an estimand-specific admission decision rather than a new universal threshold, and admission is meaningful only when the chosen reconstructor returns a finite annual metric. Autumn maintenance benefits and transported "annual peak never observed" headcounts are not retained as verified effects: the former needs an outage-cause record this archive does not publish, and the latter must stay labelled as climatological-window proxies.

Open Research Statement

Daily water-temperature records for the main retrospective panel are from nine public providers (ARSO, CHMI, FOEN, GKD Bayern, LUBW, NVE, Rijkswaterstaat, SYKE and the USGS). The independent wave-2 confirmation additionally uses hashed USGS, Hub'Eau and UK Environment Agency panels under `p0_07_wave2_ingest/wave2_inputs_manifest.csv`; the earlier sealed USGS and multi-agency transport scores remain withdrawn after audit. None required a data-use agreement. Named open licences cover the USGS (US public domain), CHMI/GKD Bayern/SYKE (CC BY 4.0), UK EA (OGL v3.0), FOEN (opendata.swiss), NVE (NLOD/CC BY), SMHI (CC BY 4.0 SE) and Hub'Eau (Licence Ouverte). ARSO and Rijkswaterstaat state attribution/source terms without naming a licence. LUBW reserves copyright: its raw values are *not* redistributed with this paper; the seven LUBW networks are available only

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through LUBW's own service, and the analysis reports derived statistics. Licences, acquisition routes and evidence are in `metadata/data_rights.csv`, `measurement_provenance_dictionary.csv` and `current_data_acquisition.csv`. Station coordinates are from those providers' catalogues, the USGS NWIS site service, and ARSO's public station feed. Daily 2 m air temperature is NASA POWER (MERRA-2) (NASA Langley Research Center, 2026). Basin characteristics are GAGES-II (Falcone, 2011).

All analysis code, derived artifacts, figures, the editable draw.io sources of the schematic figures and the frozen pre-registration are in the accompanying repository. A claim registry (`claim_registry.csv`) records each headline claim's artifact, endpoint, unit, denominator, evidence type and whether it may enter the abstract or conclusions. An automated binding check (`scripts/254_wrr_claim_binding_check.py`) recomputes the allowed headline set against live artifacts (including the wave-2 confirmation). Input-file hashes for the 146 study networks are in `results/wrr_submission_ready/current_data_manifest.c` wave-2 panel hashes are in `wave2_inputs_manifest.csv`. Source data for the figures are deposited with the analysis: the schematic figures are draw.io files (`figures/v17/drawio/`), and every data figure names its source artifact in its caption, with the plotted values also written to `figures/wrr/`. A permanent archive of the repository and a DOI will be deposited on acceptance. Certificate reproduction requires `constraints-freeze-certificate.txt` (scikit-learn 1.2.2), not the general `constraints.txt` pin.

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Your acknowledgements will appear here.

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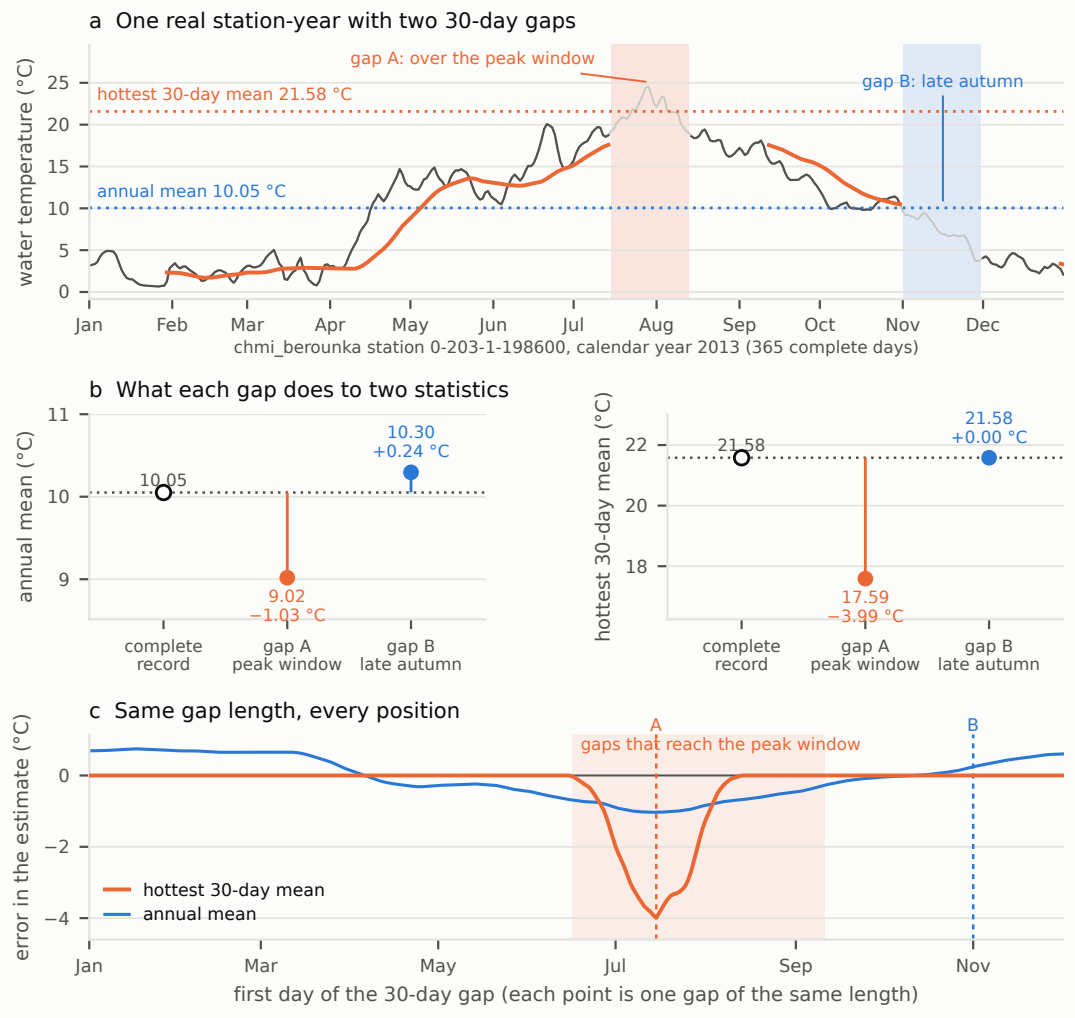


Figure 2: Two statistics, one record, two equal-length gaps. (a) A complete annual water-temperature record (CHMI Berounka, station 0-203-1-198600, 2013; n = 365 days, no missing day) with two designed 30-day gaps: gap A over the window that achieves the complete-record hottest 30-day mean (15 July to 13 August) and gap B in late autumn (1 to 30 November). The curve is dark where the gap leaves it observed and faint where the analyst would not see it. (b) Each gap's effect on each statistic, in °C: dots are estimates from the days the gap leaves, open dots and dotted lines the complete-record truth (annual mean 10.05 °C, hottest 30-day mean 21.58 °C). The panels use different vertical scales and neither starts at zero, so the printed values carry the comparison: gap A moves the hottest 30-day mean by −3.99 °C while the same length in late autumn leaves it exact, whereas the annual mean moves by −1.03 and +0.24 °C. (c) All 336 possible 30-day gap starts of the same year, showing the error each position induces in the hottest 30-day mean (orange) and the annual mean (blue); the shaded band holds starts whose window reaches the peak window. An illustration on one real record, not a result: no model, no population estimate, no test. Source data: *figures/wrr/f1_example_data.json* and *f1_example_sweep.csv*.

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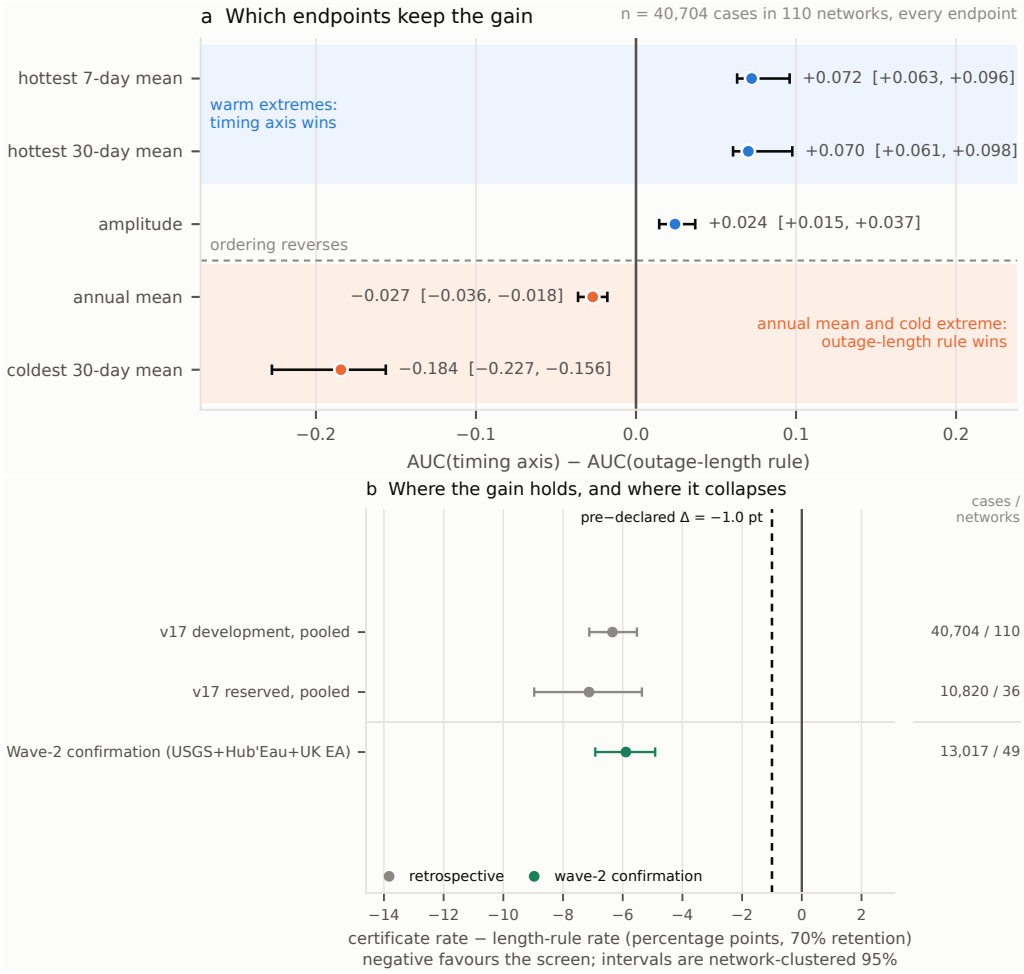


Figure 3: The same screening gain, ordered two ways. (a) Discrimination of a consequential error by endpoint, as the AUC difference between the best timing axis and the outage-length rule, with network-clustered 95% intervals; positive means the timing axis wins. Warm-season extremes favour timing; the annual mean favours length. The cold-extreme negative result in this panel uses a warm-season support axis and should not be read as "length wins for cold extremes" — cold-season support reverses that comparison (AUC ≈ 0.88 vs length 0.82; SI winter amendment). (b) Pooled contrasts for retrospective development and reserved panels (grey) and the wave-2 independent confirmation (green; USGS, Hub'Eau, UK EA; $\Delta = -5.89$ points). The first sealed USGS and multi-agency transport scores remain omitted as withdrawn after the 2026-10-13 audit. The dashed line is the pre-declared $\Delta = -1.0$ point threshold. Source data: *figures/wrr/f2_heterogeneity_data.json*.

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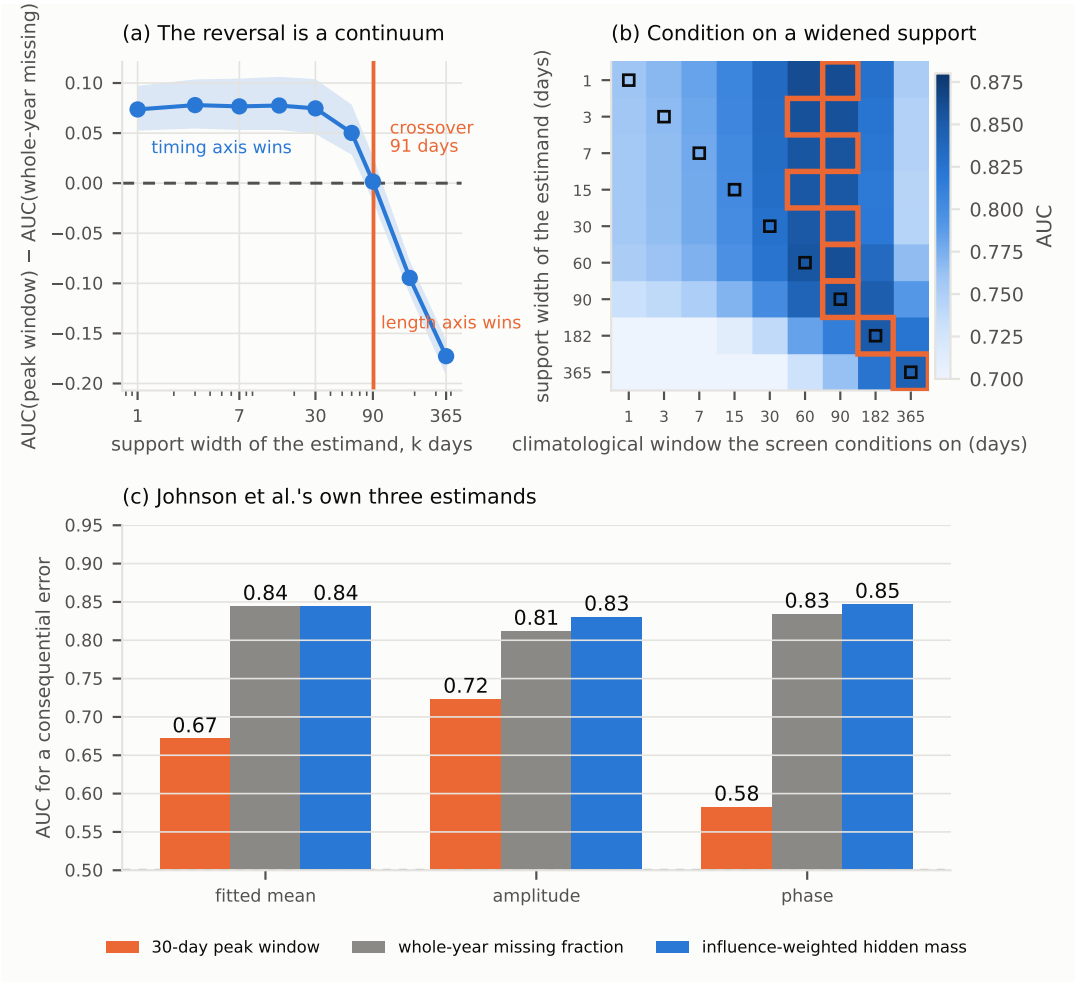


Figure 4: The transition is continuous, and it is the estimand's support that decides. Exploratory results from the opened panel, not the sealed USGS confirmation. (a) Advantage in AUC of a fixed 30-day peak window over whole-year completeness, against the support width k of the estimand (1 to 365 days), development cases; the crossing is an interval, 77 to 107 days, and the sign is reversed for $k = 182$ and $k = 365$ days. (b) AUC of each conditioning axis for each estimand, so that the reversal can be read column by column; intervals are network-clustered 95%. (c) The same axes for the three sine-fit parameters of the standing guidance, each evaluated at 0.47 of its own interannual standard deviation. Because the support of a year must be located before it can be conditioned on, the useful conditioning window is wider than the statistic itself, which is why a fixed narrow window turns over. Source data: *results/v18_closure/e6_support_width/* and *e3_sine_endpoints/*.

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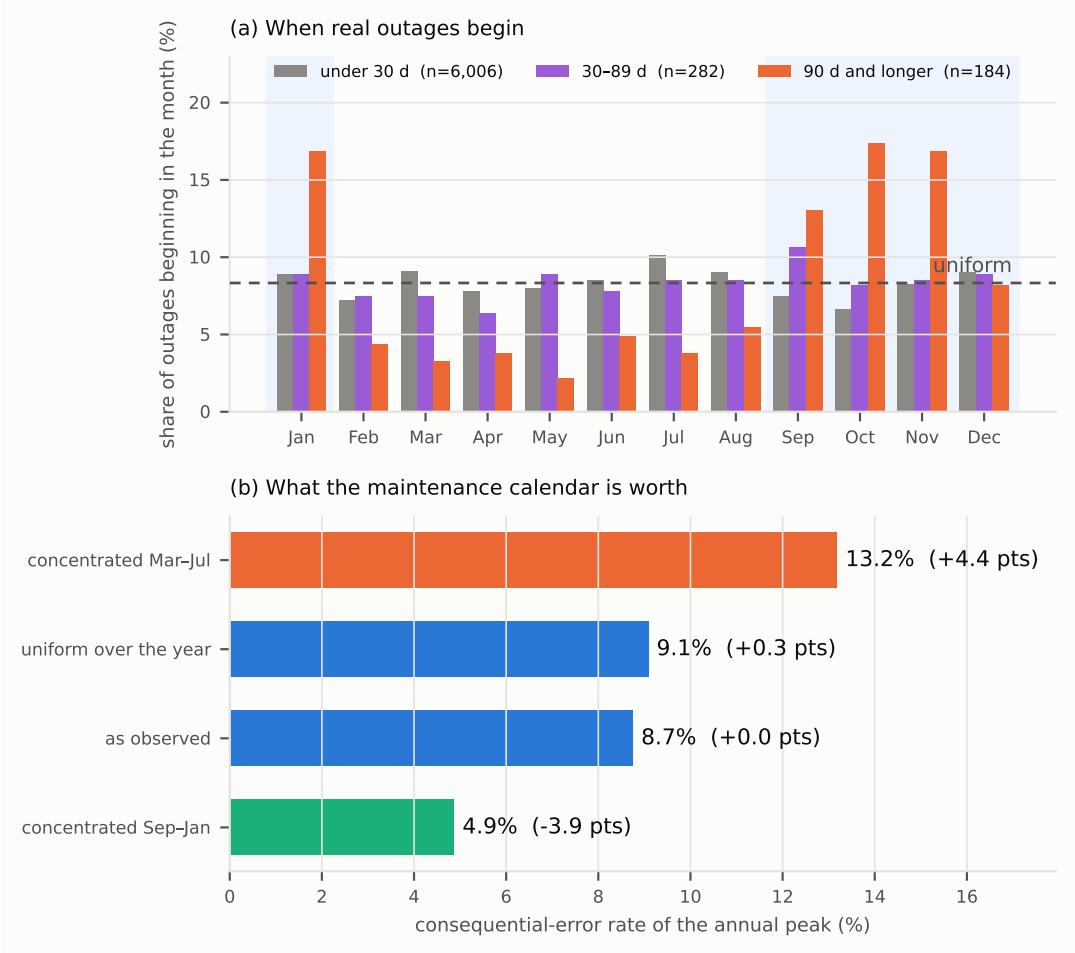


Figure 5: The gap calendar these networks actually keep. (a) Start month of the 6472 real interior gaps found in complete years, split by length class: 5246 gaps of 1–7 days, 760 of 8–29 days, 282 of 30–89 days and 184 of 90 days or longer (median gap 2 days, 95th percentile 43 days). Long outages begin in autumn and winter, which is where the peak window is not; 55.8% of them ([50.9, 60.9]%, 99 network clusters) hide less than half of the climatological peak window. (b) Exploratory maintenance-calendar counterfactual: consequential-error rate for 32 240 outage placements drawn from the observed calendar, the maintenance calendar, and a uniform calendar after the multi-year placement correction of the amendment; differences are within the interval, so no maintenance benefit is claimed. Descriptive and exploratory, not a sealed result. Source data: *results/v18_closure/e1_outage_seasonality/*.

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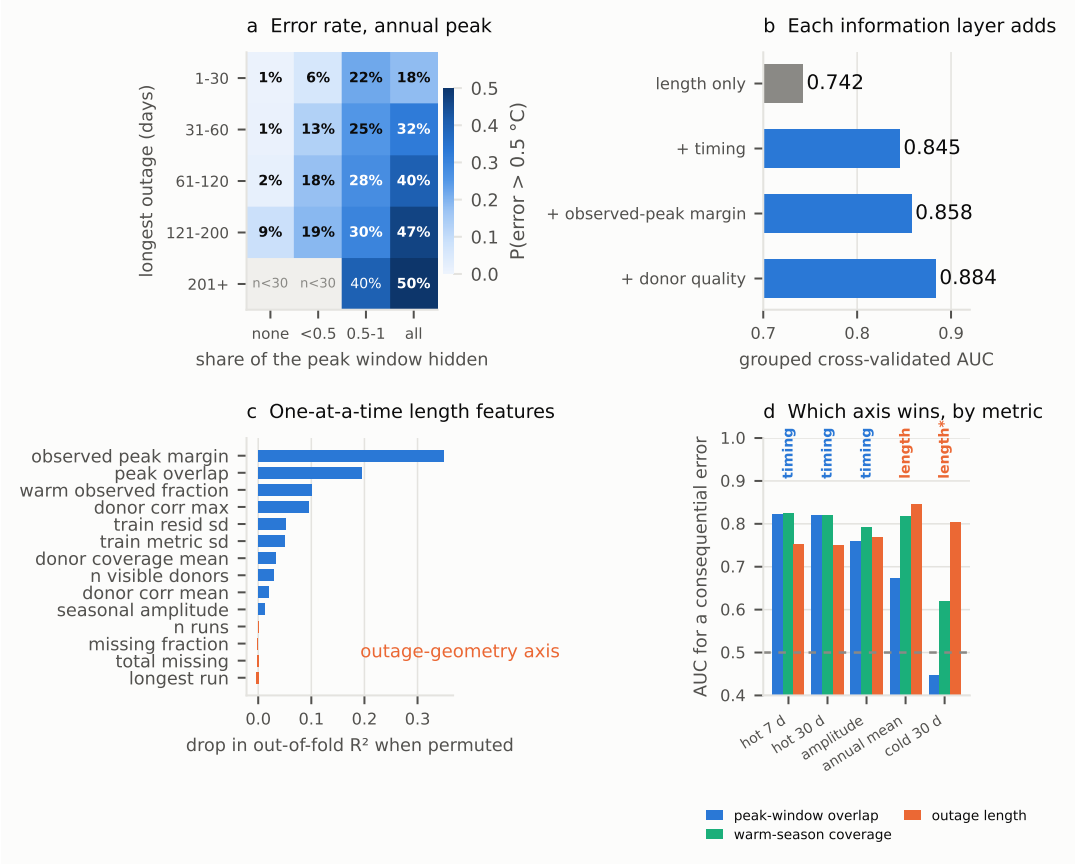


Figure 6: Position and length both move the error, and the reconstructor sets the scale. Development cases (n = 40 704 cases, 110 networks); error bars are network-clustered 95% intervals unless a panel says otherwise. (a) Growth of consequential error along each axis with the other averaged out, computed in bins of both properties: the hottest 30-day mean grows 9.6× along peak-window overlap and 3.5× along missing days, while the annual mean reverses the two (1.5× and 10.2×). (b) Out-of-fold AUC of nested feature sets grouped by network, from outage length alone (0.742) through support (0.845) and observed-peak margin (0.858) to donor quality (0.884). (c) Permutation importance of the decision-time features; once the support is in the model, outage length adds nothing measurable (−0.004 against 0.194 for peak-window overlap). (d) The winning axis by endpoint, the reversal of the main figure computed on the same cases. Source data: *results/v17_fitness/step5_mechanism/* and *step22_robustness/*.

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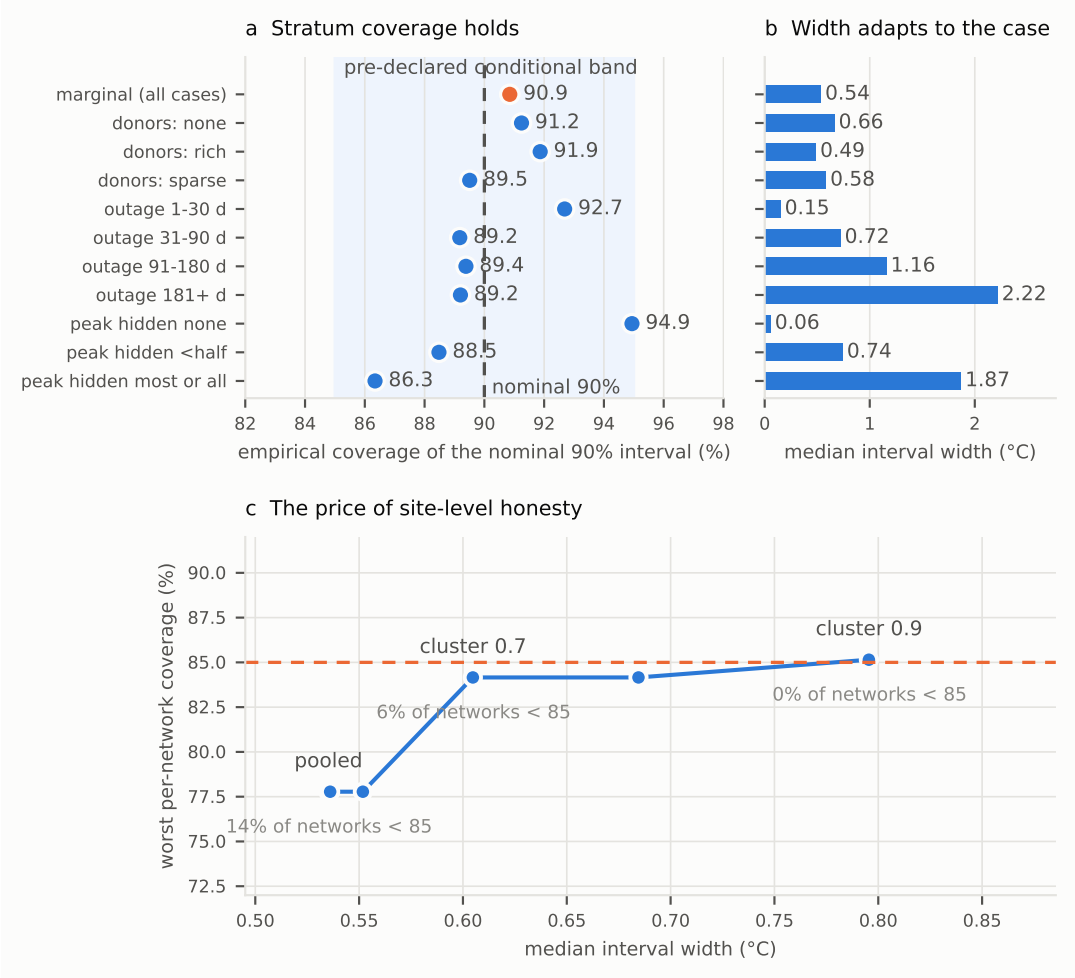


Figure 7: Is the predicted error bound honest enough to rank station-years? (a) Coverage of the nominal 90% interval by donor-regime stratum on the reserved panel: 90.9% marginally, inside the pre-declared 0.87–0.93 band, and 86.3–94.9% within every pre-declared stratum; the dashed line is the nominal level. (b) Interval width in °C against the share of the peak window the gap hides, from 0.06 °C when the window is untouched to 1.87 °C when most of it is hidden, with network-clustered bands, so the bound tracks the quantity it is meant to bound. (c) Width price of a site-level rather than pooled guarantee: the site-level quantiles reach no network below 0.85 at a 1.5-fold widening, but that level was chosen on the opened reserved panel and stays exploratory. Development cases unless a panel says otherwise. Source data: *results/v17_fitness/step8_conformal/* and *amendments/e6_support_width_uncertainty/*.

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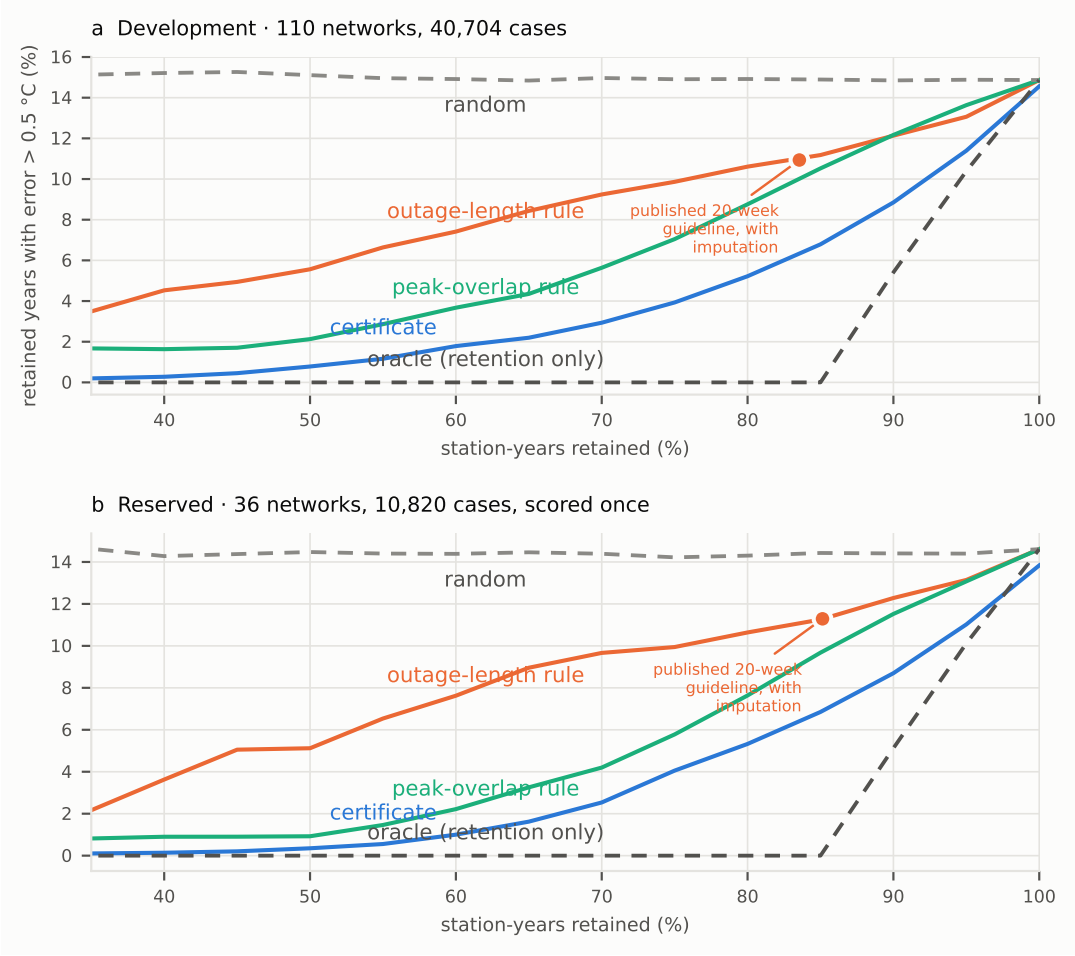


Figure 8: What equal retention costs under each admission rule. (a) Development cases (n = 40 704 cases, 110 networks): the share of retained station-years whose hottest 30-day mean is wrong by more than 0.5 °C, against the share of station-years retained, for random retention (grey), the gap-length rule (orange), peak-window overlap alone (blue) and the calibrated screen (black). (b) The same curves on the 36-network reserved panel (10 820 cases), which was opened once after the configuration was frozen. Curves are network-clustered means with 95% bands; the square marks the operating point quoted in the text, 70% retention, where the screen leaves 2.5% against 9.7% for the length rule, a contrast of −7.13 points ([−8.97, −5.36]). Source data: `results/v17_fitness/step4_ceiling/` and `step10_confirmation/`.

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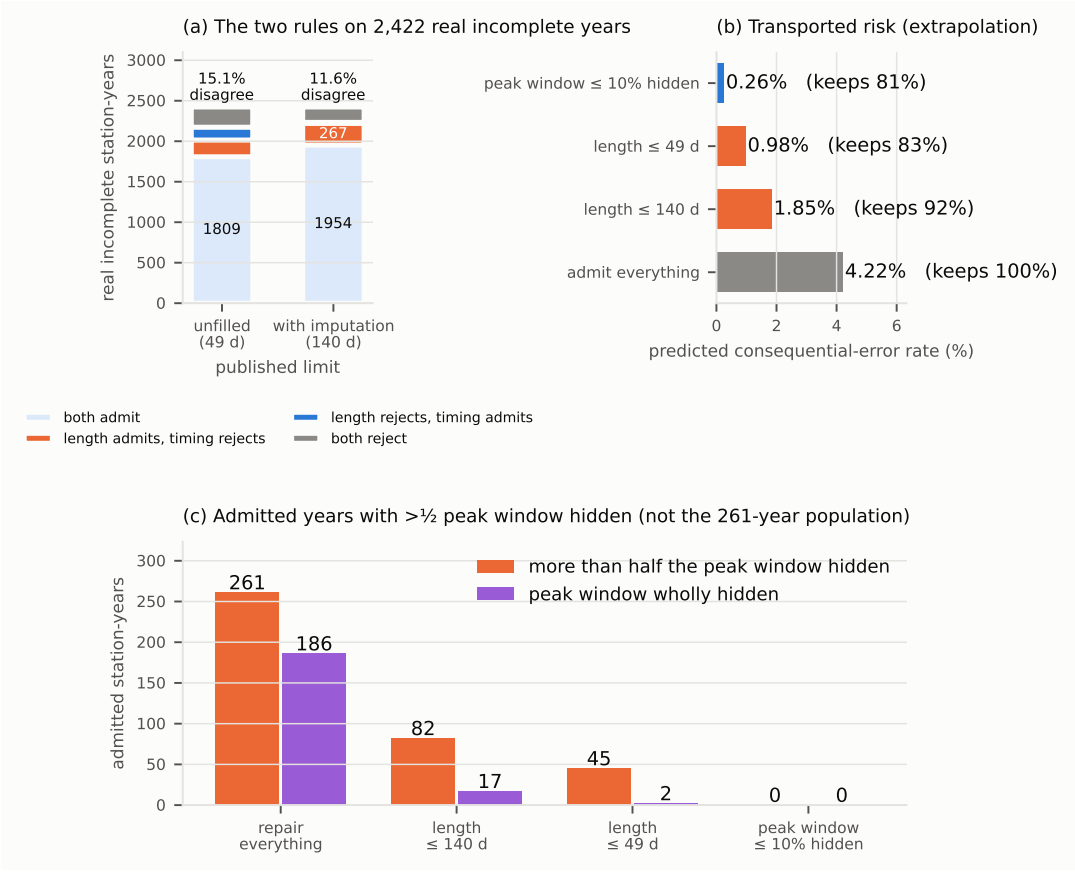


Figure 9: What the two admission rules do to records that really have gaps. (a) The length rule and the support screen applied to the 2422 incomplete station-years of this archive that have never been decided (of 7536 station-years in total, 5114 of them complete): they disagree on 15.1% of admissions ([13.4, 16.8]%, 136 network clusters). Among all 2422 years, 261 have more than half of the climatological peak window hidden; of those, the length rule at 49 d admits 45 and at 140 d admits 82 (Table 11) — not 261. (b) Predicted, not observed, consequential error for each admitted set (there is no truth in these years); the screen's admitted set is the more conservative one, and the prediction is miscalibrated by at most 6.2 points in its worst decile. (c) Incomplete years with more than half of the climatological peak window hidden (population n = 261); length-rule admits in that population are 45 or 82 depending on the threshold. 28.7% of the 2422 years hide at least one day of the climatological peak window (median 9 missing days, 90th percentile 128). "Peak window" here is the climatological window from the station's earlier years, a proxy: the positive predictive value of that proxy for a wholly unobserved realised peak is 0.74 on designed cases. Source data: *results/v18_closure/e2_archive_deployment/*.

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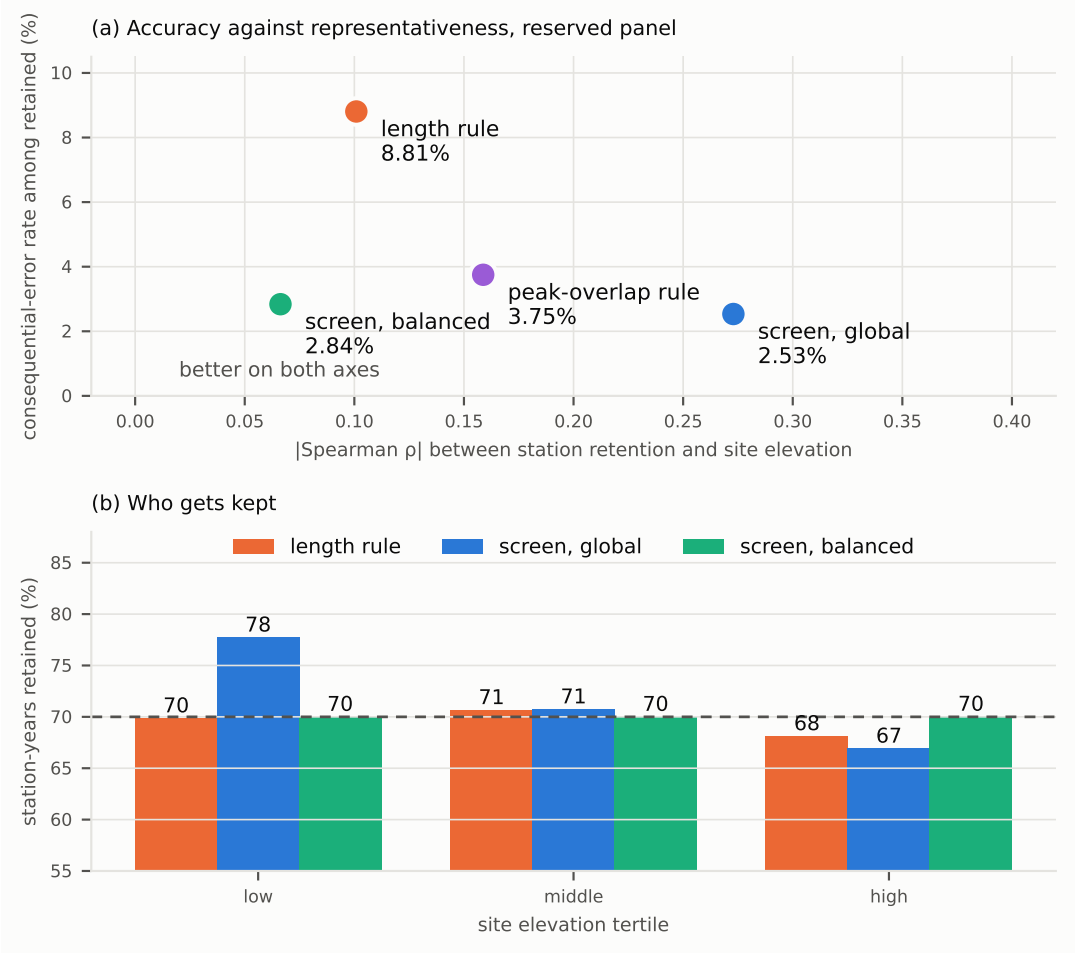


Figure 10: Which station-years the screen costs, and what balancing them would cost. (a) Across the development networks, the higher the correlation between a station's retention and its site elevation, the larger the consequential-error rate under the calibrated screen ($\rho = -0.259$, $p = 0.011$, 43 stations with GAGES-II attributes); the screen appears to select on difficulty, which tracks catchment type. (b) On the reserved panel, retention by elevation tertile under the length rule, the calibrated screen and the screen with thresholds fitted within strata: the screen retains 10.8 points fewer high-elevation station-years than low-elevation ones ($[-15.4, -0.3]$, 36 network clusters), and balancing on elevation recovers that at a cost of 0.31 points of consequential error. Bars are network-clustered 95% intervals. Source data: *results/v18_closure/e4_representativeness/* and *amendments/faithful_baseline/*.

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